Toward Robust Hyperspectral Anomaly Detection: A Quantile-Regression-Guided Joint Low-Rank and Sparse Representation Framework

Yunhang Wang, Zhaoting Liu, Yinan Zhao, Xiaorong Xu, and Hengguang Mou

***Abstract*—Hyperspectral anomaly detection (HAD) aims to identify anomalous targets against complex, noisy backgrounds, a critical yet challenging task in remote sensing. Conventional methods, particularly those relying on least-square (LS) regression, are often sensitive to heavy-tailed noise and outliers due to their implicit Gaussian noise assumption, leading to performance degradation. To address this, we propose a robust quantile-regression framework with joint low-rank and sparse representation (QR-JLRS). Our key innovation is to model the conditional quantiles of the observed hyperspectral data, decomposing them into a low-rank background, a sparse anomaly component, and the quantiles of the noise. This formulation inherently improves robustness against heavy-tailed noise, outliers, and other non-Gaussian disturbances. To efficiently solve the ensuing nonsmooth optimization problem, we devise a tailored saddle point primal-dual algorithm. Furthermore, we integrate a preconditioning strategy that employs adaptive diagonal step-size matrices derived from the linear operator's structure, substantially improving convergence stability and speed. Extensive experiments on multiple hyperspectral benchmarks demonstrate the superior robustness, detection accuracy, and practical utility of the proposed QR-JLRS framework compared to state-of-the-art alternatives.**

***Index Terms*—Hyperspectral anomaly detection (HAD), low-rank and sparse representation (LRASR), quantile regression (QR), saddle point optimization.**

I. Introduction

HYPERSPECTRAL imaging (HSI) captures abundant spectral information of observed scenes by measuring reflected or radiated energy across hundreds or even thousands of contiguous spectral bands, where each pixel is represented as a high-dimensional spectral vector characterizing the material properties at a specific spatial location [1], [2], [3], [4]. Leveraging this rich spectral representation, hyperspectral anomaly detection (HAD) aims to identify pixels whose spectral signatures significantly deviate from the background without requiring prior knowledge of target characteristics, making it a critical component in modern remote sensing systems [5]. Owing to its capability to reveal subtle material differences in complex environments, HAD has been widely applied in mineral exploration, environmental monitoring, and military surveillance [6], [7], [8]. However, the performance of HAD is often challenged in real-world scenarios, as hyperspectral data are frequently contaminated by non-Gaussian noise [9], [10], such as heavy-tailed noise or outliers, which can severely degrade the reliability of conventional statistical models, particularly those relying on least-square (LS) regression.

Traditional anomaly detection methods such as the Reed–Xiaoli (RX) detector [11] rely on multivariate Gaussian assumptions and second-order statistics, making them sensitive to outliers and noise contamination. To improve robustness, variants such as the RX detector with spatial augmentation (RXDS) have been proposed, which incorporate each pixel and its four-connected neighbors as background samples for more stable covariance estimation. Another extension is the fractional Fourier transform-based RX (FrFT-RX) [12], which uses fractional Fourier transform as preprocessing and selects the optimal transform order via entropy maximization for enhanced detection. In parallel, whitening-based strategies like the whitening spatial correlation filtering (WSCF) method [13] attempt to decorrelate spectral bands and emphasize spatial structure, yielding improved detection in cluttered scenes. However, these enhancements still fundamentally rely on sample covariance estimation and often struggle when the noise exhibits non-Gaussian or heavy-tailed characteristics. To address this, the random-selection-based anomaly detector (RSAD) [14] randomly selects representative background pixels to compute background statistics and reduces contamination by employing multiple random selections and fusing their respective detection results. An alternative line of research adopts a subspace decomposition view, modeling the hyperspectral scene as a combination of structured background and sparse anomaly [15], [16]. Among these, low-rank representation (LRR) methods [17], [18] leverage the global spectral correlation structure, assuming that background pixels lie in a low-dimensional subspace while anomalies are sparse deviations. Building on LRR, the low-rank and sparse representation (LRASR) method [19] imposes simultaneous low-rank and sparse constraints on the representation coefficient matrix itself, thereby achieving more accurate anomaly detection. Nonetheless, most LRR-based approaches still adopt the LS estimator, such as robust hyperspectral denoising

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(RhyDe) [20], which generally performs poorly in the presence of heavy-tailed noise.

To address these limitations, we propose a robust quantile-regression framework with joint LRASR (QR-JLRS). The core idea is to decompose the conditional quantiles of the observed hyperspectral image data into three parts: a low-rank background component, a sparse anomaly component, and the conditional quantiles of the noise. Unlike traditional LS-based methods, QR-JLRS employs quantile regression (QR) [21] to estimate this decomposition via a conditional quantile function. This provides a more complete characterization of the spectral data distribution than the conditional mean alone. To simultaneously capture the global low-rank structure of the background and the spatially sparse nature of anomalies, we incorporate joint LRASR priors into the framework. This formulation leads to a convex yet nonsmooth optimization problem that demands specialized algorithmic solutions.

To solve this problem efficiently, we recast it as a convex-concave saddle point problem and develop a tailored primal-dual hybrid gradient (PDHG) algorithm [22], [23], [24]. Furthermore, we enhance the algorithm using a diagonal preconditioning scheme, where scalar step sizes are replaced with adaptive diagonal matrices derived from the structure of the linear operator. This preconditioning improves both convergence speed and stability.

Extensive experiments are conducted on four widely used hyperspectral datasets, with anomaly detection evaluated under both clean and noisy conditions. To simulate diverse environmental challenges, we consider Gaussian noise as well as heavy-tailed distributions including Student's t noise and Cauchy noise. Results based on multiple complementary evaluation metrics consistently demonstrate the superior performance of QR-JLRS over a range of representative methods, including RX, RXDS, FrFT-RX, WSCF, LRR, LRASR, RSAD, and RhyDe.

In summary, the main contributions of this article are threefold.

1) A robust QR-JLRS is proposed for HAD.
2) To solve the resulting nonsmooth convex problem efficiently, a tailored saddle point primal-dual algorithm is developed, whose convergence is further accelerated and stabilized by a diagonal preconditioning strategy.
3) The proposed method is validated across multiple benchmark datasets under challenging noise conditions, demonstrating consistent improvements over a range of representative methods.

II. Problem Formulation of HAD

A. Anomaly Detection Model

HAD aims to identify pixels that deviate significantly from the background. A prevalent modeling approach characterizes the observed hyperspectral data $\boldsymbol{H} \in \mathbb{R}^{L\times N}$ (L spectral bands and N spatial pixels) as a superposition of three distinct components: background, anomalies, and noise [25]. Formally, the model is expressed as

$$\boldsymbol{H} = \boldsymbol{B} + \boldsymbol{E} + \boldsymbol{R} \tag{1}$$

where $\boldsymbol{B}, \boldsymbol{E}$, and $\boldsymbol{R} \in \mathbb{R}^{L\times N}$ represent the background, anomaly, and noise components, respectively. Given $\boldsymbol{H}$, the objective is to estimate $\boldsymbol{B}$ and $\boldsymbol{E}$ by exploiting their intrinsic structural properties. Specifically, the high spectral correlation allows the background $\boldsymbol{B}$ to be modeled as a low-rank matrix, while the anomaly matrix $\boldsymbol{E}$ is assumed to be column-wise sparse, reflecting the spatial sparsity of anomalous targets.

B. RhyDe for Anomaly Detection

Building on the general decomposition model in (1), the RhyDe framework [20] further assumes that the background component $\boldsymbol{B}$ lies in a low-dimensional subspace, leading to a more structured representation. Specifically, RhyDe expresses $\boldsymbol{B}$ as $\boldsymbol{B} = \boldsymbol{O}\boldsymbol{C}$, where $\boldsymbol{O} \in \mathbb{R}^{L\times p}$ ($p \ll L$) serves as an orthogonal basis for the signal subspace, and $\boldsymbol{C} \in \mathbb{R}^{p\times N}$ represents the corresponding coefficient matrix. The anomaly matrix $\boldsymbol{E}$ is modeled as column-wise sparse, reflecting the spatial sparsity of anomalous targets, while the residual $\boldsymbol{R}$ is typically assumed to be independent and identically distributed (i.i.d.) Gaussian noise.

The estimation of $\boldsymbol{C}$ and $\boldsymbol{E}$ is achieved by solving the following optimization problem:

$$\min_{\boldsymbol{C},\boldsymbol{E}} \frac{1}{2}\|\boldsymbol{O}\boldsymbol{C} + \boldsymbol{E} - \boldsymbol{H}\|_F^2 + \lambda_1 \phi(\boldsymbol{C}) + \lambda_2 \|\boldsymbol{E}\|_{2,1} \tag{2}$$

where $\|\boldsymbol{B}\|_F^2 = \mathrm{Tr}(\boldsymbol{B}\boldsymbol{B}^\top)$ denotes the squared Frobenius norm of $\boldsymbol{B}$, $\phi(\cdot)$ is a regularization term promoting self-similarity of the eigenimages (rows of $\boldsymbol{C}$), often implemented via plug-and-play denoisers such as BM3D [26] or WNNM [27]. The mixed $\ell_{2,1}$ norm $\|\boldsymbol{E}\|_{2,1}$ is defined as $\|\boldsymbol{E}\|_{2,1} = \sum_{i=1}^{N} \|\boldsymbol{e}_i\|_2$, where $\boldsymbol{e}_i$ denotes the ith column of $\boldsymbol{E}$, and $\|\cdot\|_2$ denotes the ℓ_2-norm. This norm encourages column-wise sparsity in $\boldsymbol{E}$, and λ_1 and λ_2 are positive regularization parameters that control the strength of the respective regularization terms.

After completing the optimization process, the anomaly score for the ith pixel is computed based on the corresponding column of the estimated anomaly matrix $\widehat{\boldsymbol{E}}$. Specifically, the detection score is defined as

$$r_i = \|\hat{\boldsymbol{e}}_i\|_2, \quad i = 1, \ldots, N \tag{3}$$

where $\hat{\boldsymbol{e}}_i$ is the ith column of $\widehat{\boldsymbol{E}}$. A pixel is identified as anomalous if its score exceeds a predefined threshold.

Although RhyDe improves robustness by explicitly modeling noise and exploiting self-similarity, its data fidelity term relies on the LS loss, which estimates the conditional mean of the data. This formulation remains sensitive to heavy-tailed noise and may bias the estimated background toward anomalous pixels, ultimately reducing the accuracy of anomaly separation.

III. Proposed QR Algorithm for HAD

A. QR Formulation of the Anomaly Detection Model

In the RhyDe framework, the background and anomaly components are traditionally estimated using an LS formulation, as shown in (2). Although LS regression effectively estimates conditional means, it is often susceptible to heteroscedasticity and fails to adequately characterize data

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distributions under non-Gaussian noise. To address these limitations, we incorporate QR, which estimates the conditional quantiles of the response variables rather than their means. This approach enables the model to focus on specific portions of the data distribution instead of being dominated by average behavior.

In this work, the background component is characterized by a dictionary-based representation [19], where the background matrix is factorized as $\boldsymbol{B} = \boldsymbol{DS}$. Here, $\boldsymbol{D} \in \mathbb{R}^{L\times M}$ serves as the background dictionary, with each row $\boldsymbol{d}_l$ representing the dictionary atom for the lth spectral band, and $\boldsymbol{S} \in \mathbb{R}^{M\times N}$ denotes the corresponding abundance or coefficient matrix. Consequently, for a specified quantile level $\tau \in (0,1)$, the conditional τ-quantile of the observed entry $H_{l,n}$ given $\boldsymbol{d}_l$ is formulated as

$$q^H_{\tau,l,n} = \inf\left\{h : P\left(H_{l,n} \le h \middle| \boldsymbol{d}_l\right) \ge \tau\right\}. \tag{4}$$

This probabilistic threshold effectively partitions the conditional distribution into two distinct regions: one capturing the cumulative probability τ below the τth quantile, and the other representing the complementary probability $1-\tau$ above it. Under this framework, the estimation of the background component $\boldsymbol{B}$ is no longer restricted to the conditional mean of the distribution. Instead, the model can be adaptively centered at a specific quantile, effectively isolating the background even in the presence of skewed noise. This flexibility, in turn, enhances the precision of anomaly detection by ensuring that $\boldsymbol{E}$ captures true structural deviations rather than residual non-Gaussian noise.

In the context of HAD, we adopt a QR model that explicitly models the conditional quantiles of the observed spectra. For a fixed quantile level $\tau \in (0,1)$, the τth conditional quantile of the observed spectral value $H_{l,n}$ (at band l and pixel n) is formulated as

$$q^H_{\tau,l,n} = \boldsymbol{d}_l^\top \boldsymbol{s}_n + e_{l,n} + q^R_{\tau,l,n} \tag{5}$$

where $\boldsymbol{d}_l^\top \boldsymbol{s}_n$ represents the structured background component, $e_{l,n}$ denotes the anomaly contribution, and $q^R_{\tau,l,n}$ accounts for the conditional quantile of the residual noise.

By aggregating across all spectral bands and spatial pixels, we extend the pixel-wise model to a matrix-level QR formulation

$$\boldsymbol{Q}^H_\tau = \boldsymbol{DS} + \boldsymbol{E} + \boldsymbol{Q}^R_\tau \tag{6}$$

where $\boldsymbol{Q}^H_\tau \in \mathbb{R}^{L\times N}$ represents the matrix of conditional τ-quantiles for the observed hyperspectral data. In this decomposition, $\boldsymbol{DS}$ constitutes the background component, $\boldsymbol{E} \in \mathbb{R}^{L\times N}$ denotes the anomaly matrix, and $\boldsymbol{Q}^R_\tau \in \mathbb{R}^{L\times N}$ accounts for the conditional τ-quantiles of the residual noise. Physically, $\boldsymbol{Q}^R_\tau$ represents the variations in the observed spectra arising from sensor noise and atmospheric effects such as scattering and absorption.

B. QR Framework With Joint LRASR for Anomaly Detection

Building upon the QR formulation in (6), the QR estimator for the background representation coefficients $\boldsymbol{S}$, the anomaly matrix $\boldsymbol{E}$, and the noise conditional quantile matrix $\boldsymbol{Q}^R_\tau$ can be obtained by solving the following optimization problem:

$$\left\{\boldsymbol{S},\boldsymbol{E},\boldsymbol{Q}^R_\tau\right\} = \underset{\boldsymbol{S},\boldsymbol{E},\boldsymbol{Q}^R_\tau}{\arg\min} f\left(\boldsymbol{S},\boldsymbol{E},\boldsymbol{Q}^R_\tau\right) + \lambda\varphi(\boldsymbol{S}) + \beta\psi(\boldsymbol{E}) \tag{7}$$

with

$$f\left(\boldsymbol{S},\boldsymbol{E},\boldsymbol{Q}^R_\tau\right) \triangleq \sum_{l=1}^{L}\sum_{n=1}^{N}\rho_\tau\left(h_{l,n} - \boldsymbol{d}_l^\top \boldsymbol{s}_n - e_{l,n} - q^R_{\tau,l,n}\right) \tag{8}$$

where $\varphi(\cdot)$ and $\psi(\cdot)$ are regularizers, and $\rho_\tau(\cdot)$ is the nondifferentiable quantile loss function, also known as the check function

$$\rho_\tau(r) = \begin{cases}\tau r, & r \ge 0\\ (\tau-1)r, & r < 0.\end{cases} \tag{9}$$

The regularization terms $\varphi(\cdot)$ and $\psi(\cdot)$ are designed to exploit the intrinsic structural priors of hyperspectral imagery for enhanced detection performance. Specifically, the background spectra are typically characterized by a high degree of spatiospectral correlation, as they are generated from a limited number of materials. Consequently, the background component resides in a low-dimensional subspace, motivating a low-rank constraint on the coefficient matrix $\boldsymbol{S}$ [17]. Beyond low-rankness, $\boldsymbol{S}$ also exhibits intrinsic sparsity [19], as each pixel is typically modeled as a linear combination of only a few dominant dictionary atoms. Imposing element-wise sparsity on $\boldsymbol{S}$ not only improves the interpretability of the representation but also effectively suppresses noise and mitigates the influence of redundant atoms. Meanwhile, since anomalies occur infrequently and are spatially localized, the anomaly matrix $\boldsymbol{E}$ is constrained to be column-wise sparse, reflecting its nature as a set of distinct deviations from the background subspace.

Accordingly, the nuclear norm $\|\cdot\|_*$ and the ℓ_1-norm $\|\cdot\|_1$ are jointly employed to promote the low-rank and sparse structures of $\boldsymbol{S}$, respectively. To characterize the anomalies, the $\ell_{2,1}$-norm is applied to $\boldsymbol{E}$ to enforce column-wise sparsity. Integrating these regularizers into the QR framework defined in (7) leads to the following composite optimization problem:

$$\begin{aligned}\left\{\boldsymbol{S},\boldsymbol{E},\boldsymbol{Q}^R_\tau\right\} = \underset{\boldsymbol{S},\boldsymbol{E},\boldsymbol{Q}^R_\tau}{\arg\min} & f\left(\boldsymbol{S},\boldsymbol{E},\boldsymbol{Q}^R_\tau\right) + \lambda_1\|\boldsymbol{S}\|_*\\ & + \lambda_2\|\boldsymbol{S}\|_1 + \lambda_3\|\boldsymbol{E}\|_{2,1}\end{aligned} \tag{10}$$

where $\|\cdot\|_*$ denotes the matrix nuclear norm (sum of singular values), and $\|\cdot\|_1$ represents the element-wise ℓ_1-norm (sum of absolute values of all entries). The regularization parameters λ_1, λ_2, and $\lambda_3 > 0$ serve to balance the competing objectives of data fidelity, background low-rankness, coefficient sparsity, and the spatial scarcity of anomalies.

Given that the check function $\rho_\tau(v)$ in (10) is nondifferentiable at the origin, it presents a challenge for direct gradient-based optimization. To address this nonsmoothness, we leverage a dual formulation by expressing $\rho_\tau(v)$ through its convex conjugate

$$\rho_\tau(v) = \sup_{u\in\mathbb{R}}\left\{uv - \rho^*_\tau(u)\right\} = \sup_{\tau-1\le u\le\tau} uv \tag{11}$$

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where

$$\rho_\tau^*(v) = \begin{cases} 0, & \text{if } \tau - 1 \le v < \tau \\ \infty, & \text{otherwise.} \end{cases} \tag{12}$$

This reformulation facilitates the transformation of the original nonsmooth problem into a saddle point problem, which can be efficiently solved using proximal-point algorithms or primal-dual methods. To this end, we transform the fidelity term $f(\boldsymbol{S}, \boldsymbol{E}, \boldsymbol{Q}_\tau^R)$ into its variational dual form by introducing the auxiliary matrix $\boldsymbol{Y} \in \mathbb{R}^{L\times N}$. Leveraging the conjugate property of the check function, the formulation in (8) can be rewritten as

$$\begin{aligned} f\left(\boldsymbol{S}, \boldsymbol{E}, \boldsymbol{Q}_\tau^R\right) &= \sum_{l=1}^{L}\sum_{n=1}^{N} \max_{\tau-1\le y_{l,n}<\tau} y_{l,n} \\ &\quad \left(h_{l,n} - \boldsymbol{d}_l^\top \boldsymbol{s}_n - e_{l,n} - q_{\tau,l,n}^R\right) \\ &= \max_{\boldsymbol{Y}\in\mathbb{R}^{L\times N}} \left\{\langle \boldsymbol{Y}, \boldsymbol{H} - \boldsymbol{D}\boldsymbol{S} - \boldsymbol{E} - \boldsymbol{Q}_\tau^R\rangle - \mathcal{I}_{\mathbb{Y}}(\boldsymbol{Y})\right\} \end{aligned} \tag{13}$$

where $\langle \boldsymbol{U}, \boldsymbol{V}\rangle = \mathrm{Tr}(\boldsymbol{U}^\top \boldsymbol{V})$ denotes the Frobenius inner product, $\mathbb{Y} = \{\boldsymbol{Y} \in \mathbb{R}^{L\times N} : \tau - 1 \le y_{l,n} \le \tau, \forall l, n\}$ is the convex constraint set for the dual variables, and $\mathcal{I}_{\mathbb{Y}}(\boldsymbol{Y})$ is the indicator function

$$\mathcal{I}_{\mathbb{Y}}(\boldsymbol{Y}) = \begin{cases} 0, & \text{if } y_{l,n} \in [\tau - 1, \tau] \quad \forall l, n \\ +\infty, & \text{otherwise.} \end{cases} \tag{14}$$

To achieve a unified primal-dual optimization framework, we similarly express the regularization terms—namely, the nuclear norm, the ℓ_1-norm, and the $\ell_{2,1}$-norm—through their respective dual norm representations.

1) For the nuclear norm of $\boldsymbol{S}$

$$\|\boldsymbol{S}\|_* = \max_{\|\boldsymbol{Z}_1\|_\sigma \le 1} \langle \boldsymbol{Z}_1, \boldsymbol{S}\rangle \tag{15}$$

where $\|\boldsymbol{Z}_1\|_\sigma$ denotes the spectral norm (the largest singular value) of the dual variable $\boldsymbol{Z}_1$.

2) For the element-wise ℓ_1-norm of $\boldsymbol{S}$

$$\|\boldsymbol{S}\|_1 = \max_{\|\boldsymbol{Z}_2\|_\infty \le 1} \langle \boldsymbol{Z}_2, \boldsymbol{S}\rangle \tag{16}$$

where $\|\boldsymbol{Z}_2\|_\infty$ represents the element-wise ℓ_∞-norm (maximum absolute entry) of $\boldsymbol{Z}_2$.

3) For the $\ell_{2,1}$-norm of the anomaly matrix $\boldsymbol{E}$

$$\|\boldsymbol{E}\|_{2,1} = \max_{\boldsymbol{Z}_3\in\mathbb{Z}_3} \langle \boldsymbol{Z}_3, \boldsymbol{E}\rangle \tag{17}$$

where $\mathbb{Z}_3 = \{\boldsymbol{Z}_3 : \|\boldsymbol{Z}_3(:, n)\|_2 \le 1 \forall n\}$, meaning each column of the dual matrix $\boldsymbol{Z}_3$ is constrained within a unit ℓ_2-ball.

Combining these expressions, we reformulate the original problem (10) as a saddle point optimization

$$\begin{aligned} \min_{\boldsymbol{S},\boldsymbol{E},\boldsymbol{Q}_\tau^R} \max_{\boldsymbol{Y},\boldsymbol{Z}_1,\boldsymbol{Z}_2,\boldsymbol{Z}_3} &\left\{\left\langle \boldsymbol{Y}, \boldsymbol{H} - \boldsymbol{D}\boldsymbol{S} - \boldsymbol{E} - \boldsymbol{Q}_\tau^R\right\rangle - \mathcal{I}_{\mathbb{Y}}(\boldsymbol{Y})\right. \\ &+ \lambda_1 \langle \boldsymbol{Z}_1, \boldsymbol{S}\rangle + \lambda_2 \langle \boldsymbol{Z}_2, \boldsymbol{S}\rangle + \lambda_3 \langle \boldsymbol{Z}_3, \boldsymbol{E}\rangle \\ &\left. - \mathcal{I}_{\mathbb{Z}_1}(\boldsymbol{Z}_1) - \mathcal{I}_{\mathbb{Z}_2}(\boldsymbol{Z}_2) - \mathcal{I}_{\mathbb{Z}_3}(\boldsymbol{Z}_3)\right\} \end{aligned} \tag{18}$$

where each indicator function imposes norm constraints corresponding to the regularization dual domains

$$\mathcal{I}_{\mathbb{Z}_1}(\boldsymbol{Z}_1) = \begin{cases} 0, & \text{if } \|\boldsymbol{Z}_1\|_\sigma \le 1 \\ +\infty, & \text{otherwise} \end{cases} \tag{19}$$

$$\mathcal{I}_{\mathbb{Z}_2}(\boldsymbol{Z}_2) = \begin{cases} 0, & \text{if } \|\boldsymbol{Z}_2\|_\infty \le 1 \\ +\infty, & \text{otherwise} \end{cases} \tag{20}$$

$$\mathcal{I}_{\mathbb{Z}_3}(\boldsymbol{Z}_3) = \begin{cases} 0, & \text{if } \|\boldsymbol{Z}_3(:, n)\|_2 \le 1 \quad \forall n \\ +\infty, & \text{otherwise.} \end{cases} \tag{21}$$

To solve the saddle point optimization problem in (18), we employ the PDHG algorithm, also known as the Chambolle–Pock method. This approach iteratively updates the primal variables $\{\boldsymbol{S}, \boldsymbol{E}, \boldsymbol{Q}_\tau^R\}$ and the dual variables $\{\boldsymbol{Y}, \boldsymbol{Z}_1, \boldsymbol{Z}_2, \boldsymbol{Z}_3\}$ by alternating between gradient ascent/descent and proximal mapping steps. The overall iterative scheme is summarized as follows.

Primal Updates:

$$\boldsymbol{S}^{(\ell+1)} = \boldsymbol{S}^{(\ell)} + \mu\left(\boldsymbol{D}^\top \boldsymbol{Y}^{(\ell)} - \lambda_1 \boldsymbol{Z}_1^{(\ell)} - \lambda_2 \boldsymbol{Z}_2^{(\ell)}\right) \tag{22a}$$

$$\boldsymbol{E}^{(\ell+1)} = \boldsymbol{E}^{(\ell)} + \mu\left(\boldsymbol{Y}^{(\ell)} - \lambda_3 \boldsymbol{Z}_3^{(\ell)}\right) \tag{22b}$$

$$\boldsymbol{Q}_\tau^{R(\ell+1)} = \boldsymbol{Q}_\tau^{R(\ell)} + \mu \boldsymbol{Y}^{(\ell)}. \tag{22c}$$

Extrapolation:

$$\widetilde{\boldsymbol{S}}^{(\ell+1)} = 2\boldsymbol{S}^{(\ell+1)} - \boldsymbol{S}^{(\ell)} \tag{23a}$$

$$\widetilde{\boldsymbol{E}}^{(\ell+1)} = 2\boldsymbol{E}^{(\ell+1)} - \boldsymbol{E}^{(\ell)} \tag{23b}$$

$$\widetilde{\boldsymbol{Q}}_\tau^{R(\ell+1)} = 2\boldsymbol{Q}_\tau^{R(\ell+1)} - \boldsymbol{Q}_\tau^{R(\ell)}. \tag{23c}$$

Dual Updates via Proximal Projections:

$$\begin{aligned} \boldsymbol{Y}^{(\ell+1)} = \mathrm{prox}_{\eta\mathcal{I}_{\mathbb{Y}}}\Bigg(\boldsymbol{Y}^{(\ell)} - \eta\Bigg(&\boldsymbol{D}\widetilde{\boldsymbol{S}}^{(\ell+1)} \\ &+ \widetilde{\boldsymbol{E}}^{(\ell+1)} + \widetilde{\boldsymbol{Q}}_\tau^{R(\ell+1)} - \boldsymbol{H}\Bigg)\Bigg) \end{aligned} \tag{24a}$$

$$\boldsymbol{Z}_1^{(\ell+1)} = \mathrm{prox}_{\eta\mathcal{I}_{\mathbb{Z}_1}}\left(\boldsymbol{Z}_1^{(\ell)} + \eta\lambda_1 \widetilde{\boldsymbol{S}}^{(\ell+1)}\right) \tag{24b}$$

$$\boldsymbol{Z}_2^{(\ell+1)} = \mathrm{prox}_{\eta\mathcal{I}_{\mathbb{Z}_2}}\left(\boldsymbol{Z}_2^{(\ell)} + \eta\lambda_2 \widetilde{\boldsymbol{S}}^{(\ell+1)}\right) \tag{24c}$$

$$\boldsymbol{Z}_3^{(\ell+1)} = \mathrm{prox}_{\eta\mathcal{I}_{\mathbb{Z}_3}}\left(\boldsymbol{Z}_3^{(\ell)} + \eta\lambda_3 \widetilde{\boldsymbol{E}}^{(\ell+1)}\right). \tag{24d}$$

In this scheme, (23a)–(23c) represent the extrapolation steps that ensure the convergence of the primal-dual iterations. μ and η are positive primal and dual step sizes, respectively. The operator $\mathrm{prox}_{\eta h}(\cdot)$ denotes the proximal operator associated with the indicator function h, which in this context simplifies to a Euclidean projection onto the respective convex constraint sets $\mathbb{Y}$ and $\mathbb{Z}_i$. Specifically, for a function $h(\cdot)$, the proximal operator is defined as

$$\mathrm{prox}_{\eta h}(\boldsymbol{x}) = \arg\min_{\boldsymbol{y}} \left\{h(\boldsymbol{y}) + \frac{1}{2\eta}\|\boldsymbol{y} - \boldsymbol{x}\|^2\right\}. \tag{25}$$

Each dual variable is updated using a proximal operator corresponding to its constraint set. These proximal mappings are derived based on the indicator functions associated with the feasible domains $\mathbb{Y}$ and $\mathbb{Z}_k$ for $k = 1, 2, 3$. Specifically, the proximal operator $\mathrm{prox}_{\eta\mathcal{I}_{\mathbb{Y}}}(\boldsymbol{Y})$ projects $\boldsymbol{Y}$ onto the interval $[\tau - 1, \tau]$ element-wise, which yields

$$\mathrm{prox}_{\eta\mathcal{I}_{\mathbb{Y}}}(\boldsymbol{Y}) = \min(\tau, \max(\tau - 1, \boldsymbol{Y})). \tag{26}$$

Similarly, the proximal operators for the spectral, ℓ_∞, and ℓ_2 norm constraints are obtained as closed-form projections

$$(\text{Spectral}): \quad \mathrm{prox}_{\eta\mathcal{I}_{\mathbb{Z}_1}}(\boldsymbol{Z}_1) = \boldsymbol{U}\mathrm{diag}(\min(\boldsymbol{\sigma}, 1))\boldsymbol{V}^\top \tag{27}$$

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Algorithm 1 QR-JLRS

Input: Data matrix $\boldsymbol{H}$, dictionary matrix $\boldsymbol{D}$, parameters $\lambda_1, \lambda_2, \lambda_3, \mu, \eta, \tau$, tolerance ε, maximum iteration MaxIter.
Initialize primal variable $\boldsymbol{S}^{(0)}$, $\boldsymbol{E}^{(0)}$, $\boldsymbol{Q}_\tau^{R(0)}$, dual variables $\boldsymbol{Y}^{(0)}, \boldsymbol{Z}_1^{(0)}, \boldsymbol{Z}_2^{(0)}, \boldsymbol{Z}_3^{(0)}$, set $\ell = 0$.
while $\frac{\|\boldsymbol{H}-\boldsymbol{D}\boldsymbol{S}^{(\ell)}-\boldsymbol{E}^{(\ell)}-\boldsymbol{Q}_\tau^{R(\ell)}\|_{\mathrm{F}}}{\|\boldsymbol{H}\|_{\mathrm{F}}} > \varepsilon$ **or** $\ell <$ MaxIter **do**
  Update variable $\boldsymbol{S}^{(\ell+1)}$, $\boldsymbol{E}^{(\ell+1)}$, $\boldsymbol{Q}_\tau^{R(\ell+1)}$ using (22a), (22b) and (22c)
  Update variable $\widetilde{\boldsymbol{S}}^{(\ell+1)}$, $\widetilde{\boldsymbol{E}}^{(\ell+1)}$, $\widetilde{\boldsymbol{Q}}_\tau^{R(\ell+1)}$ using (23a), (23b) and (23c)
  Update dual variable $\boldsymbol{Y}^{(\ell+1)}$ using (24a) and (26)
  Update dual variable $\boldsymbol{Z}_1^{(\ell+1)}$ using (24b) and (27)
  Update dual variable $\boldsymbol{Z}_2^{(\ell+1)}$ using (24c) and (28)
  Update dual variable $\boldsymbol{Z}_3^{(\ell+1)}$ using (24d) and (29)
  Increment $\ell = \ell + 1$.
end while
Output: $\boldsymbol{S}^{(\ell+1)}$, $\boldsymbol{E}^{(\ell+1)}$, $\boldsymbol{Q}_\tau^{R(\ell+1)}$.

$$(\ell_\infty): \quad \mathrm{prox}_{\eta\mathcal{I}_{\mathbb{Z}_2}}(\boldsymbol{Z}_2) = \min(1, \max(-1, \boldsymbol{Z}_2)) \tag{28}$$

$$(\ell_2): \mathrm{prox}_{\eta\mathcal{I}_{\mathbb{Z}_3}}(\boldsymbol{Z}_3(:,n)) = \frac{\boldsymbol{Z}_3(:,n)}{\max(1, \|\boldsymbol{Z}_3(:,n)\|_2)} \quad \forall n \tag{29}$$

where $\boldsymbol{Z}_1 = \boldsymbol{U}\mathrm{diag}(\boldsymbol{\sigma})\boldsymbol{V}^\top$ denotes the singular value decomposition (SVD) of $\boldsymbol{Z}_1$. Among these, the projection onto the spectral norm ball is less trivial, and its derivation is provided in the Appendix. The remaining proximal mappings correspond to standard element-wise or column-wise Euclidean projections and are therefore omitted for brevity.

With all primal and dual update steps derived above, we now summarize the proposed algorithm under the saddle point optimization framework. The resulting procedure, referred to as the QR-JLRS, is detailed in Algorithm 1.

C. Convergence Guarantee

To facilitate the convergence analysis, we define the concatenated primal and dual variables as $\mathcal{X} = [\boldsymbol{S}^\top, \boldsymbol{E}^\top, (\boldsymbol{Q}_\tau^R)^\top]^\top$ and $\mathcal{Y} = [\boldsymbol{Y}^\top, \boldsymbol{Z}_1^\top, \boldsymbol{Z}_2^\top, \boldsymbol{Z}_3^\top]^\top$, respectively. The problem in (18) can then be compactly expressed in the standard saddle point form

$$\min_{\mathcal{X}} \max_{\mathcal{Y}} \{\langle \mathcal{K}\mathcal{X}, \mathcal{Y}\rangle + G(\mathcal{X}) - F(\mathcal{Y})\} \tag{30}$$

where $G(\mathcal{X}) = 0$ and $F(\mathcal{Y}) = \mathcal{I}_{\mathbb{U}}(\mathcal{Y}) - \langle \boldsymbol{Y}, \boldsymbol{H}\rangle$. The linear operator $\mathcal{K}$ is defined as in (31). The term $\mathcal{I}_{\mathbb{U}}(\mathcal{Y})$ denotes the indicator function over the Cartesian product $\mathbb{U} = \mathbb{Y} \times \mathbb{Z}_1 \times \mathbb{Z}_2 \times \mathbb{Z}_3$, which decomposes as $\mathcal{I}_{\mathbb{U}}(\mathcal{Y}) = \mathcal{I}_{\mathbb{Y}}(\boldsymbol{Y}) + \sum_{i=1}^3 \mathcal{I}_{\mathbb{Z}_i}(\boldsymbol{Z}_i)$. The linear operator $\mathcal{K}$ is defined such that

$$\mathcal{K} = \begin{bmatrix} -\boldsymbol{D} & -\boldsymbol{I}_L & -\boldsymbol{I}_L \\ \lambda_1 \boldsymbol{I}_M & \boldsymbol{0}_{M\times L} & \boldsymbol{0}_{M\times L} \\ \lambda_2 \boldsymbol{I}_M & \boldsymbol{0}_{M\times L} & \boldsymbol{0}_{M\times L} \\ \boldsymbol{0}_{L\times M} & \lambda_3 \boldsymbol{I}_L & \boldsymbol{0}_{L\times L} \end{bmatrix} \in \mathbb{R}^{(2L+2M)\times(2L+M)} \tag{31}$$

with $\boldsymbol{I}_M$ and $\boldsymbol{I}_L$ denote identity matrices of dimensions $M \times M$ and $L \times L$, respectively.

The saddle point problem (30) can be efficiently solved using the PDHG algorithm with the following iterative update scheme for primal and dual variables:

$$\begin{cases} \mathcal{X}^{(\ell+1)} = \mathcal{X}^{(\ell)} - \mu\mathcal{K}^\top\mathcal{Y}^{(\ell)} \\ \widetilde{\mathcal{X}}^{(\ell+1)} = 2\mathcal{X}^{(\ell+1)} - \mathcal{X}^{(\ell)} \\ \mathcal{Y}^{(\ell+1)} = \mathrm{prox}_{\eta\mathcal{I}_{\mathbb{U}}(\cdot)}\left(\mathcal{Y}^{(\ell)} + \eta\left(\mathcal{K}\widetilde{\mathcal{X}}^{(\ell+1)} + \mathcal{H}\right)\right) \end{cases} \tag{32}$$

where $\mathcal{H} = [\boldsymbol{H}^\top, \boldsymbol{0}_{M\times N}^\top, \boldsymbol{0}_{M\times N}^\top, \boldsymbol{0}_{L\times N}^\top]^\top$ is an extended matrix accounting for the linear term $\langle \boldsymbol{Y}, \boldsymbol{H}\rangle$.

According to [28], the convergence of the PDHG algorithm is guaranteed if the step sizes satisfy the condition

$$\mu\eta\|\mathcal{K}\|^2 < 1 \tag{33}$$

where $\|\mathcal{K}\|$ denotes the spectral norm of the composite operator $\mathcal{K}$. Under this condition, the sequence $\{(\mathcal{X}^{(\ell)}, \mathcal{Y}^{(\ell)})\}_{\ell=0,1,\ldots}$ generated by (32) converges to the optimal saddle point $(\mathcal{X}^*, \mathcal{Y}^*)$ of problem (30) with a convergence rate of $O(1/\ell)$. It is worth noting that the iterative scheme in (32) is equivalent to that in (22)–(24). Therefore, the condition in (33) strictly guarantees the convergence of (22)–(24) to the optimal solution of the proposed saddle point optimization problem in (18).

D. Nonscalar Step Sizes via Diagonal Preconditioning

To enhance the convergence speed and stability of the proposed saddle point optimization algorithm, we introduce a preconditioned variant of the PDHG method. This approach replaces the scalar step sizes μ and η with symmetric positive-definite matrices (also referred to as preconditioners) $\boldsymbol{\Sigma}$ and $\boldsymbol{\Gamma}$, respectively. The iterative updates are given by

$$\begin{cases} \mathcal{X}^{(\ell+1)} = \mathcal{X}^{(\ell)} - \boldsymbol{\Sigma}\mathcal{K}^\top\mathcal{Y}^{(\ell)} \\ \widetilde{\mathcal{X}}^{(\ell+1)} = 2\mathcal{X}^{(\ell+1)} - \mathcal{X}^{(\ell)} \\ \mathcal{Y}^{(\ell+1)} = \mathrm{prox}_{\boldsymbol{\Gamma},\mathcal{I}_{\mathbb{U}}(\cdot)}\left(\mathcal{Y}^{(\ell)} + \boldsymbol{\Gamma}\left(\mathcal{K}\widetilde{\mathcal{X}}^{(\ell+1)} + \mathcal{H}\right)\right) \end{cases} \tag{34}$$

where $\mathrm{prox}_{\boldsymbol{\Gamma},\mathcal{I}_{\mathbb{U}}(\cdot)}(\boldsymbol{x}) \triangleq \arg\min_{\boldsymbol{y}} 1/2\langle \boldsymbol{x}-\boldsymbol{y}, \boldsymbol{\Gamma}(\boldsymbol{x}-\boldsymbol{y})\rangle + \mathcal{I}_{\mathbb{U}}(\boldsymbol{y})$. Following [29, Lemma 1], if the preconditioners $\boldsymbol{\Sigma}$ and $\boldsymbol{\Gamma}$ satisfy the condition

$$\left\|\boldsymbol{\Sigma}^{\frac{1}{2}}\mathcal{K}\boldsymbol{\Gamma}^{\frac{1}{2}}\right\|^2 < 1 \tag{35}$$

then the sequence $\{(\mathcal{X}^{(\ell)}, \mathcal{Y}^{(\ell)})\}_{\ell=0,1,\ldots}$ is guaranteed to converge.

Appropriate preconditioners not only accelerate convergence but also avoid the tedious tuning of scalar step sizes. However, their design depends strongly on the structure of the linear operator $\mathcal{K}$. As shown in (31), the operator $\mathcal{K}$ admits an explicit block structure. This explicit representation allows us to directly evaluate the row and column norms of $\mathcal{K}$, thereby enabling the construction of efficient diagonal preconditioners. Specifically, we define $\boldsymbol{\Sigma} = \mathrm{diag}(\mu_1, \ldots, \mu_{2L+M})$ and $\boldsymbol{\Gamma} = \mathrm{diag}(\eta_1, \ldots, \eta_{2L+2M})$ with the diagonal elements computed as

$$\mu_j = \frac{1}{\sum_{i=1}^{2L+2M}|\mathcal{K}_{i,j}|^{2-\alpha}}, \quad \eta_i = \frac{1}{\sum_{j=1}^{2L+M}|\mathcal{K}_{i,j}|^{\alpha}}. \tag{36}$$

It has been proven that such a construction guarantees the convergence condition in (35) is satisfied for any $\alpha \in [0, 2]$,

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and hence allows the iterations (34) convergence to the optimal solution of the problem (30).

Furthermore, by inspecting the structure of the operator $\mathcal{K}$ in (31), the diagonal elements of the preconditioners in (36) can be explicitly decomposed according to the different components of the primal and dual variables. This decomposition provides a clear block-wise correspondence, which is particularly convenient when performing matrix-based updates. Specifically, the entries of the primal preconditioner $\boldsymbol{\Sigma}$ can be written as

$$\mu_j = \begin{cases} \dfrac{1}{(\lambda_1)^{2-\alpha} + (\lambda_2)^{2-\alpha} + \sum_{i=1}^{L} |\mathcal{K}_{i,j}|^{2-\alpha}}, & j = 1, \ldots, M \\ \dfrac{1}{1 + (\lambda_3)^{2-\alpha}}, & j = M+1, \ldots, L+M \\ 1, & j = L+M+1, \ldots, 2L+M \end{cases} \tag{37}$$

where the first, second, and third blocks correspond to the primal variables $\boldsymbol{S}, \boldsymbol{E}, \boldsymbol{Q}_\tau^R$, respectively, which can be written in matrix form as

$$\begin{aligned} \boldsymbol{\Lambda}_{\mu,\boldsymbol{S}} &\triangleq \operatorname{diag}(\mu_1, \ldots, \mu_M) \\ \boldsymbol{\Lambda}_{\mu,\boldsymbol{E}} &\triangleq \operatorname{diag}(\mu_{M+1}, \ldots, \mu_{L+M}) \\ \boldsymbol{\Lambda}_{\mu,\boldsymbol{Q}_\tau^R} &\triangleq \operatorname{diag}(\mu_{L+M+1}, \ldots, \mu_{2L+M}). \end{aligned} \tag{38}$$

Similarly, the entries of the dual preconditioner $\boldsymbol{\Gamma}$ are

$$\eta_i = \begin{cases} \dfrac{1}{\sum_{j=1}^{2L+M} |\mathcal{K}_{i,j}|^{\alpha}}, & i = 1, \ldots, L \\ \dfrac{1}{(\lambda_1)^{\alpha}}, & i = L+1, \ldots, L+M \\ \dfrac{1}{(\lambda_2)^{\alpha}}, & i = (L+M) \\ & \quad +1, \ldots, (L+M)+M \\ \dfrac{1}{(\lambda_3)^{\alpha}}, & i = (L+2M) \\ & \quad +1, \ldots, (L+2M)+L \end{cases} \tag{39}$$

where the four blocks correspond to the dual variables $\boldsymbol{Y}, \boldsymbol{Z}_1, \boldsymbol{Z}_2, \boldsymbol{Z}_3$, respectively, which can be expressed as

$$\begin{aligned} \boldsymbol{\Lambda}_{\eta,\boldsymbol{Y}} &\triangleq \operatorname{diag}(\eta_1, \ldots, \eta_L) \\ \boldsymbol{\Lambda}_{\eta,\boldsymbol{Z}_1} &\triangleq \operatorname{diag}(\eta_{L+1}, \ldots, \eta_{L+M}) \\ \boldsymbol{\Lambda}_{\eta,\boldsymbol{Z}_2} &\triangleq \operatorname{diag}(\eta_{L+M+1}, \ldots, \eta_{L+2M}) \\ \boldsymbol{\Lambda}_{\eta,\boldsymbol{Z}_3} &\triangleq \operatorname{diag}(\eta_{L+2M+1}, \ldots, \eta_{2L+2M}). \end{aligned} \tag{40}$$

With this explicit correspondence between the diagonal blocks of the preconditioners and the primal/dual variables, the iterations (34) can be reformulated as follows.

Primal Updates:

$$\boldsymbol{S}^{(\ell+1)} = \boldsymbol{S}^{(\ell)} + \boldsymbol{\Lambda}_{\mu,\boldsymbol{S}} \left(\boldsymbol{D}^\top \boldsymbol{Y}^{(\ell)} - \lambda_1 \boldsymbol{Z}_1^{(\ell)} - \lambda_2 \boldsymbol{Z}_2^{(\ell)} \right) \tag{41a}$$

$$\boldsymbol{E}^{(\ell+1)} = \boldsymbol{E}^{(\ell)} + \boldsymbol{\Lambda}_{\mu,\boldsymbol{E}} \left(\boldsymbol{Y}^{(\ell)} - \lambda_3 \boldsymbol{Z}_3^{(\ell)} \right) \tag{41b}$$

$$\boldsymbol{Q}_\tau^{R(\ell+1)} = \boldsymbol{Q}_\tau^{R(\ell)} + \boldsymbol{\Lambda}_{\mu,\boldsymbol{Q}_\tau^R} \boldsymbol{Y}^{(\ell)}. \tag{41c}$$

Extrapolation:

$$\widetilde{\boldsymbol{S}}^{(\ell+1)} = 2\boldsymbol{S}^{(\ell+1)} - \boldsymbol{S}^{(\ell)} \tag{42a}$$

$$\widetilde{\boldsymbol{E}}^{(\ell+1)} = 2\boldsymbol{E}^{(\ell+1)} - \boldsymbol{E}^{(\ell)} \tag{42b}$$

Algorithm 2 Preconditioned QR-JLRS

Input: Data matrix $\boldsymbol{H}$, dictionary matrix $\boldsymbol{D}$, parameters $\lambda_1, \lambda_2, \lambda_3, \tau, \alpha$, tolerance ε, maximum iteration MaxIter.
Initialize primal variable $\boldsymbol{S}^{(0)}$, $\boldsymbol{E}^{(0)}$, $\boldsymbol{Q}_\tau^{R(0)}$, dual variables $\boldsymbol{Y}^{(0)}, \boldsymbol{Z}_1^{(0)}, \boldsymbol{Z}_2^{(0)}, \boldsymbol{Z}_3^{(0)}$, set $\ell = 0$.
while $\frac{\|\boldsymbol{H} - \boldsymbol{D}\boldsymbol{S}^{(\ell)} - \boldsymbol{E}^{(\ell)} - \boldsymbol{Q}_\tau^{R(\ell)}\|_{\mathrm{F}}}{\|\boldsymbol{H}\|_{\mathrm{F}}} > \varepsilon$ **or** $\ell <$ MaxIter **do**
  Update variable $\boldsymbol{S}^{(\ell+1)}$, $\boldsymbol{E}^{(\ell+1)}$, $\boldsymbol{Q}_\tau^{R(\ell+1)}$ using (41a), (41b) and (41c)
  Update variable $\widetilde{\boldsymbol{S}}^{(\ell+1)}$, $\widetilde{\boldsymbol{E}}^{(\ell+1)}$, $\widetilde{\boldsymbol{Q}}_\tau^{R(\ell+1)}$ using (42a), (42b) and (42c)
  Update dual variable $\boldsymbol{Y}^{(\ell+1)}$ using (43a) and (26)
  Update dual variable $\boldsymbol{Z}_1^{(\ell+1)}$ using (43b) and (27)
  Update dual variable $\boldsymbol{Z}_2^{(\ell+1)}$ using (43c) and (28)
  Update dual variable $\boldsymbol{Z}_3^{(\ell+1)}$ using (43d) and (29)
  Increment $\ell = \ell + 1$.
end while
Output: $\boldsymbol{S}^{(\ell+1)}$, $\boldsymbol{E}^{(\ell+1)}$, $\boldsymbol{Q}_\tau^{R(\ell+1)}$.

$$\widetilde{\boldsymbol{Q}}_\tau^{R(\ell+1)} = 2\boldsymbol{Q}_\tau^{R(\ell+1)} - \boldsymbol{Q}_\tau^{R(\ell)}. \tag{42c}$$

Dual Updates via Proximal Projections:

$$\boldsymbol{Y}^{(\ell+1)} = \operatorname{prox}_{\mathcal{I}_{\mathbb{Y}}} \left(\boldsymbol{Y}^{(\ell)} - \boldsymbol{\Lambda}_{\eta,\boldsymbol{Y}} \left(\boldsymbol{D}\widetilde{\boldsymbol{S}}^{(\ell+1)} + \widetilde{\boldsymbol{E}}^{(\ell+1)} + \widetilde{\boldsymbol{Q}}_\tau^{R(\ell+1)} - \boldsymbol{H} \right) \right) \tag{43a}$$

$$\boldsymbol{Z}_1^{(\ell+1)} = \operatorname{prox}_{\mathcal{I}_{\mathbb{Z}_1}} \left(\boldsymbol{Z}_1^{(\ell)} + \lambda_1 \boldsymbol{\Lambda}_{\eta,\boldsymbol{Z}_1} \widetilde{\boldsymbol{S}}^{(\ell+1)} \right) \tag{43b}$$

$$\boldsymbol{Z}_2^{(\ell+1)} = \operatorname{prox}_{\mathcal{I}_{\mathbb{Z}_2}} \left(\boldsymbol{Z}_2^{(\ell)} + \lambda_2 \boldsymbol{\Lambda}_{\eta,\boldsymbol{Z}_2} \widetilde{\boldsymbol{S}}^{(\ell+1)} \right) \tag{43c}$$

$$\boldsymbol{Z}_3^{(\ell+1)} = \operatorname{prox}_{\mathcal{I}_{\mathbb{Z}_3}} \left(\boldsymbol{Z}_3^{(\ell)} + \lambda_3 \boldsymbol{\Lambda}_{\eta,\boldsymbol{Z}_3} \widetilde{\boldsymbol{E}}^{(\ell+1)} \right). \tag{43d}$$

Finally, we adopt diagonal preconditioning by replacing scalar step sizes with adaptive diagonal matrices. This leads to a preconditioned version of the QR-JLRS algorithm; the full iterative procedure is outlined in Algorithm 2.

IV. Experiments Results

A. Experimental Settings

1) Datasets Description: To comprehensively evaluate the performance of the proposed QR-JLRS framework, we conduct experiments on four representative hyperspectral datasets: AVIRIS-1 [30], airport, beach, and urban [31]. These datasets are commonly adopted in the field for validating HAD algorithms, and they span a variety of real-world scenes—ranging from structured urban areas to natural coastal and airport environments—offering diverse spatial structures, background complexities, and anomaly patterns.

The AVIRIS-1 dataset is a 100×100 subimage cropped from a larger AVIRIS scene acquired over San Diego, as shown in Fig. 1. Fig. 1(a) presents the full hyperspectral image, while Fig. 1(b) highlights the selected region used for anomaly detection experiments. The subimage contains 189 spectral bands and includes three aircraft targets, which are treated as ground-truth anomalies. In addition, the airport–beach–urban (ABU) is a composite dataset constructed from 13 publicly available AVIRIS scenes. In this study, we select one representative

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(a) (b)

Fig. 1. AVIRIS San Diego scene and cropped AVIRIS-1 subimage. (a) Full San Diego hyperspectral image. (b) AVIRIS-1 subimage used for anomaly detection experiments, containing three aircraft as anomalies.



(a) (b) (c)

Fig. 2. Representative hyperspectral scenes from the ABU dataset. (a) Airport scene. (b) Beach scene. (c) Urban scene.

image from each of the airport, beach, and urban categories for anomaly detection experiments. These scenes, illustrated in Fig. 2, exhibit distinct spatial layouts and spectral characteristics. Specifically, Fig. 2(a)–(c) displays the selected images from the airport, beach, and urban categories, respectively. Prior to analysis, corrupted spectral bands with low signal-to-noise ratios (SNRs) or strong atmospheric absorption are removed to ensure data quality.

2) Benchmark Methods: To benchmark detection performance, the proposed QR-JLRS is compared against nine representative anomaly detectors: the classical RX detector [11], its spatial variant (RXDS), the FrFT-RX [12], WSCF [13], LRR [17], LRASR [19], RSAD [14], and RhyDe [20]. In addition, to explicitly evaluate the contribution of the QR fidelity term, we further include an LS-based variant of the proposed model, termed LS-JLRS, as an ablation baseline. This variant replaces the QR loss with a standard LS formulation while preserving the same joint regularization terms, enabling a fair comparison to isolate the effect of the loss function. The specific formulation of LS-JLRS is given by

$$\{\boldsymbol{S},\boldsymbol{E}\} = \arg\min_{\boldsymbol{S},\boldsymbol{E}} \frac{1}{2}\|\boldsymbol{H}-\boldsymbol{D}\boldsymbol{S}-\boldsymbol{E}\|_F^2 + \lambda_1\|\boldsymbol{S}\|_* + \lambda_2\|\boldsymbol{S}\|_1 + \lambda_3\|\boldsymbol{E}\|_{2,1}. \tag{44}$$

All methods are implemented under a unified preprocessing pipeline and evaluated under four conditions: clean (noise-free) data, and three noisy scenarios with additive Gaussian noise ($\mu = 0$, $\sigma^2 = 0.01$), Student's t noise (two degrees of freedom), and Cauchy noise. For all noisy cases, the SNR is fixed at 35 dB, defined as $\mathrm{SNR} = 10\log_{10}\mathbb{E}[\boldsymbol{H}^\top\boldsymbol{H}]/\mathbb{E}[\boldsymbol{R}^\top\boldsymbol{R}]$, where $\boldsymbol{R}$ denotes the artificially introduced noise matrix, and $\mathbb{E}[\cdot]$ denotes the expectation operator. This design enables a fair and comprehensive evaluation of robustness under various noise environments.

3) Dictionary Construction: In this work, the background dictionary $\boldsymbol{D}$ is constructed following the strategy proposed in [19]. The core idea is to build a representative background dictionary by clustering hyperspectral pixels and selecting reliable background samples within each cluster. Specifically, K-means clustering is first employed to partition the data into K clusters so that different background material classes can be effectively captured. Then, within each cluster, a preselection strategy based on the Mahalanobis distance is applied, where pixels with smaller distances are more likely to correspond to background components rather than anomalies. Consequently, a subset of P pixels with the smallest Mahalanobis distances is selected from each cluster to form the dictionary atoms. In our implementation, the number of clusters is set to $K = 15$, and the maximum number of K-means iterations is 200. For each cluster, we select $P = 25$ pixels with the smallest Mahalanobis distances as background atoms. The final dictionary $\boldsymbol{D}$ is obtained by aggregating the selected atoms from all valid clusters. It is worth noting that all methods requiring a background dictionary, including QR-JLRS, LS-JLRS, LRR, and LRASR, utilize the same constructed dictionary $\boldsymbol{D}$ to ensure a fair comparison.

4) Performance Evaluation Metrics: For performance evaluation, the receiver operating characteristic (ROC) curve and the area under the ROC curve (AUC) are commonly employed as metrics. The ROC curve illustrates the tradeoff between the probability of false alarm P_{FA} (x-axis) and the probability of detection P_D (y-axis) across different decision thresholds. Specifically, given a threshold applied to the detector's test statistic, the detection and false alarm probabilities, respectively, can be estimated as $P_D = N_{\mathrm{TT}}/N_T$ and $P_{\mathrm{FA}} = N_{\mathrm{BT}}/N_B$, where N_{TT} denotes the number of anomaly target pixels correctly identified, N_T is the total number of anomaly pixels, N_{BT} is the number of background pixels incorrectly labeled as targets, and N_B represents the total number of background pixels. By varying the threshold over the range of the test statistic, a series of (P_D, P_{FA}) pairs can be obtained to generate the ROC curve.

The comprehensive experimental results are reported through multiple complementary evaluation metrics, providing a thorough assessment of the proposed QR-JLRS framework's effectiveness and robustness in diverse HAD settings.

B. Convergence Behavior Analysis of QR-JLRS and Preconditioned QR-JLRS

This experiment investigates the convergence behavior of the proposed QR-JLRS algorithm and its preconditioned variant, under four different noise conditions: noise-free, Gaussian noise, Student's t noise, and Cauchy noise. The *relative reconstruction error* at the ℓth iteration is evaluated over 200 iterations on four hyperspectral datasets (AVIRIS-1, airport, beach, and urban), defined as $(\|\boldsymbol{H}-\boldsymbol{D}\boldsymbol{S}^{(\ell)}-\boldsymbol{E}^{(\ell)}-\boldsymbol{Q}_\tau^{R(\ell)}\|_{\mathrm{F}})/\|\boldsymbol{H}\|_{\mathrm{F}}$.

The experimental results are illustrated in Fig. 3. Specifically, the preconditioned QR-JLRS achieves a steady-state

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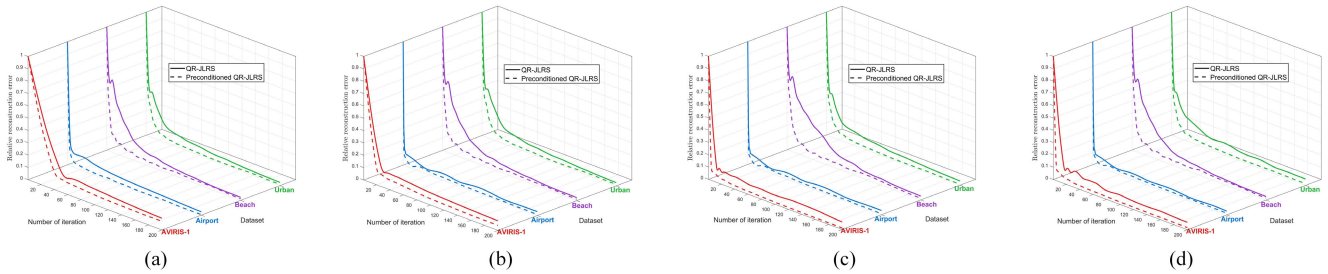


Fig. 3. Convergence curves of relative reconstruction error for QR-JLRS and preconditioned QR-JLRS on four datasets under four different noise conditions. (a) Noise-free. (b) Gaussian noise. (c) Student's t noise. (d) Cauchy noise.

response within fewer iterations compared to the baseline, which employs scalar step sizes. Beyond acceleration, the preconditioned approach exhibits superior stability; while the conventional QR-JLRS suffers from slight oscillations—particularly in the presence of stochastic noise—the integration of preconditioners effectively dampens these fluctuations, yielding a smoother convergence curve. Furthermore, the preconditioned algorithm yields a lower final relative reconstruction error across all noise regimes.

Given this performance advantage, the preconditioned QR-JLRS is employed in the subsequent comparative experiments against other state-of-the-art algorithms. For consistency and fair comparison, the LS-JLRS variant is also implemented using the same preconditioned primal-dual framework in all subsequent experiments.

C. Parameter Sensitivity Analysis

Although the preconditioned QR-JLRS algorithm removes the scalar step sizes μ and η by introducing diagonal preconditioners, the model still involves several important parameters, including λ_1, λ_2, λ_3, τ, and α. Therefore, a comprehensive parameter sensitivity analysis is conducted to evaluate the influence of these parameters on detection performance.

Before presenting the experimental results, the roles of the regularization parameters are further clarified based on the optimization model in (10). The parameters λ_1, λ_2, and λ_3 are the regularization parameters associated with the nuclear norm $\|\boldsymbol{S}\|_*$, the ℓ_1-norm $\|\boldsymbol{S}\|_1$, and the $\ell_{2,1}$-norm $\|\boldsymbol{E}\|_{2,1}$, respectively. Specifically, λ_1 controls the strength of the low-rank constraint on $\boldsymbol{S}$, λ_2 controls the sparsity level of the coefficient matrix $\boldsymbol{S}$, and λ_3 regulates the magnitude of the anomaly component $\boldsymbol{E}$ through column-wise sparsity.

A larger λ_1 leads to a more compact background representation but may suppress fine details, while a larger λ_2 encourages sparser representations at the cost of reduced flexibility. The parameter λ_3 plays a critical role in anomaly detection: a large value suppresses the anomaly term and may lead to missed detections, whereas a small value increases sensitivity but may introduce false alarms by capturing noise. Therefore, proper selection of these parameters is essential to balance background modeling accuracy and anomaly detection performance.

Regarding the quantile parameter τ, as shown in (9), it determines the asymmetry of the quantile loss. When $\tau = 0.5$, the loss becomes symmetric and reduces to the absolute loss. When $\tau > 0.5$, positive residuals are penalized more heavily, whereas when $\tau < 0.5$, negative residuals receive stronger penalties. This property enables the model to flexibly adapt to different data distributions. It is worth noting that the parameter α is fixed to 1 in all experiments, since it has been proven that this construction guarantees that the convergence condition in (35) is satisfied for any $\alpha \in [0, 2]$, ensuring convergence of the iterative scheme in (34).

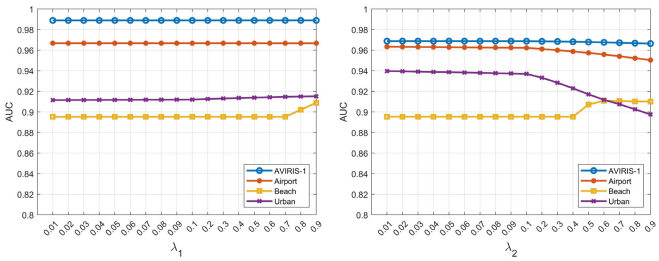


Fig. 4. Sensitivity analysis of the regularization parameters λ_1 and λ_2 on AUC performance across different hyperspectral datasets under the noise-free setting. (a) λ_1 varies. (b) λ_2 varies.

The analysis is first conducted under $\lambda_1 = \lambda_2 = \lambda_3 = 0.01$, $\tau = 0.5$, and $\alpha = 1$. Under this setting, λ_1 and λ_2 are varied independently, as shown in Fig. 4. The results demonstrate stable performance over $\lambda_1 \in [0.01, 0.7]$ and $\lambda_2 \in [0.01, 0.1]$. Accordingly, $\lambda_1 = \lambda_2 = 0.05$ are adopted. With λ_1 and λ_2 fixed, the effects of λ_3 and τ are further analyzed, as shown in Fig. 5. The results indicate that τ has a more pronounced impact, while stable and high performance is achieved for $\lambda_3 \in [0.2, 0.5]$. Thus, $\lambda_3 = 0.5$ is selected.

Regarding the quantile parameter τ, it is observed that for most datasets (AVIRIS-1, airport, and urban), the performance remains relatively stable within a broad range of $\tau \in [0.1, 0.9]$. Considering that background pixels dominate the scene, a moderately small value (e.g., $\tau = 0.3$) is adopted to impose a heavier penalty on negative residuals, thereby encouraging a tighter background fit and improving the separability between anomalies and the background while maintaining detection sensitivity. For the beach dataset, however, the optimal performance is achieved when τ takes very small values (e.g., $\tau \in [0.0001, 0.001]$). This is attributed to the homogeneous background with low spectral variability and sparse but

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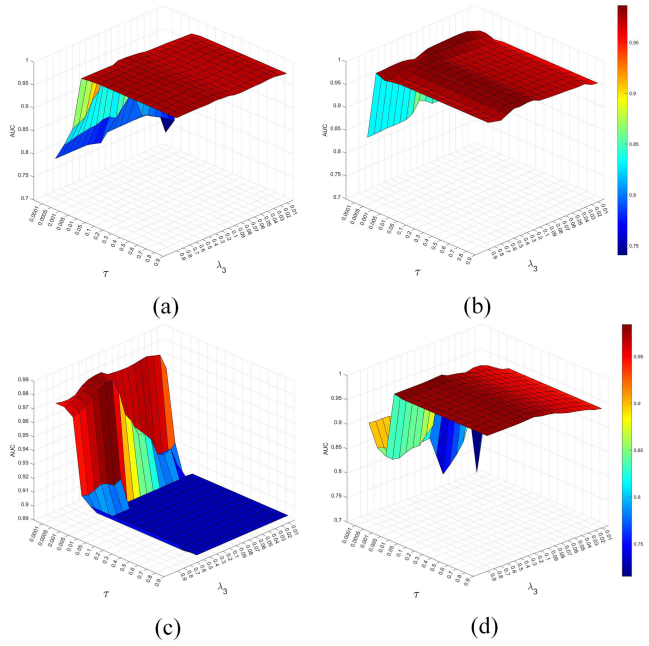


Fig. 5. Joint sensitivity analysis of the quantile parameter τ and regularization parameter λ_3 on AUC performance under different hyperspectral datasets in the noise-free setting. (a) AVIRIS-1. (b) Airport. (c) Beach. (d) Urban.

high-intensity anomalies. In this case, even a moderately small τ may still allow high-valued anomalies to influence background estimation, thereby weakening the background–anomaly contrast. Using an extremely small τ alleviates this effect by enforcing a more accurate background fit and reducing the influence of anomalies on background estimation. Based on this observation, $\tau = 0.001$ is selected for this dataset.

These results demonstrate that the proposed method is robust to a reasonably wide range of parameter settings, while also highlighting the importance of properly selecting τ to adapt to different data characteristics.

Finally, regarding the implementation of comparison algorithms, all competing methods are implemented according to their original literature. Parameters are either set using recommended values or tuned within reasonable ranges to ensure their best performance under the same experimental settings.

D. Visual Comparison of Anomaly Detection Maps

Figs. 6–9 present the anomaly detection heat maps obtained by different methods under noise-free conditions, Gaussian noise, t-distribution noise, and Cauchy noise, respectively.

In the noise-free case, most detectors are able to highlight anomalous targets to some extent. However, the proposed QR-JLRS produces more concentrated anomaly responses with clearer spatial structures and reduced background interference. This demonstrates that the proposed QR framework with joint low-rank and sparse enables more accurate anomaly detection even in ideal scenarios.

When Gaussian noise is introduced, conventional methods such as LRR and RX produce heat maps with blurred artifacts. In contrast, methods like RhyDe and the proposed QR-JLRS, which incorporate noise modeling directly into their decomposition frameworks, maintain stable anomaly saliency and preserve target integrity. This demonstrates their enhanced robustness against Gaussian noise perturbations.

More importantly, under heavy-tailed noise conditions including t-distribution noise and Cauchy noise, the superiority of QR-JLRS becomes particularly evident. Traditional anomaly detection methods such as RX, RXDS, and WSCF tend to misidentify a large number of noise outliers as anomalies, manifesting as scattered speckles throughout the heat maps. Meanwhile, methods relying on LS regression, such as RhyDe and LS-JLRS, suffer from performance degradation due to their sensitivity to outliers, resulting in increased false detections and blurred anomaly responses. By contrast, QR-JLRS consistently delivers well-localized anomalies with significantly suppressed background responses, highlighting the inherent robustness of QR to heavy-tailed noise. This indicates that QR-JLRS can effectively suppress the interference of heavy-tailed noise in anomaly detection, thereby improving detection accuracy.

E. ROC Curve Analysis and AUC Comparison

The quantitative detection performance is further evaluated using ROC curves and the corresponding AUC values, as illustrated in Figs. 10–13 and summarized in Table I, where the best results in each scenario are highlighted in bold. As illustrated in Figs. 10–13, under both noise-free and noisy conditions, QR-JLRS consistently achieves ROC curves closer to the upper left corner across all datasets, indicating a favorable tradeoff between detection probability and false alarm rate. In particular, under noise-free conditions, the ROC curves of QR-JLRS lie markedly above those of the competing methods. This observation is corroborated by the AUC values reported in Table I, where QR-JLRS attains the best performance on all four datasets.

In the presence of Gaussian noise, although several methods maintain reasonable detection capability, QR-JLRS demonstrates improved stability and still achieves the highest AUC values across all datasets. For example, on the beach dataset under Gaussian noise, QR-JLRS attains an AUC of 0.9719, while traditional methods such as RX and WSCF drop to 0.8363 and 0.8183, respectively. This confirms the robustness of QR-JLRS to Gaussian noise pollution.

Under t-distribution noise and Cauchy noise, the performance gap becomes more pronounced. The LS-based RhyDe detector experiences significant degradation due to outliers; on the beach dataset under t-distribution noise, QR-JLRS achieves an AUC of 0.9606, while RhyDe drops to an AUC of 0.7309. Similarly, on the airport dataset under Cauchy noise, RhyDe suffers a severe performance drop to an AUC of 0.7451, whereas QR-JLRS maintains a high AUC of 0.9735. In contrast, methods like LRR and RX show inconsistent performance across different noise types. These results clearly demonstrate the advantage of QR over LS regression in heavy-tailed noise environments, as QR-JLRS consistently yields superior and stable AUC values in most cases.

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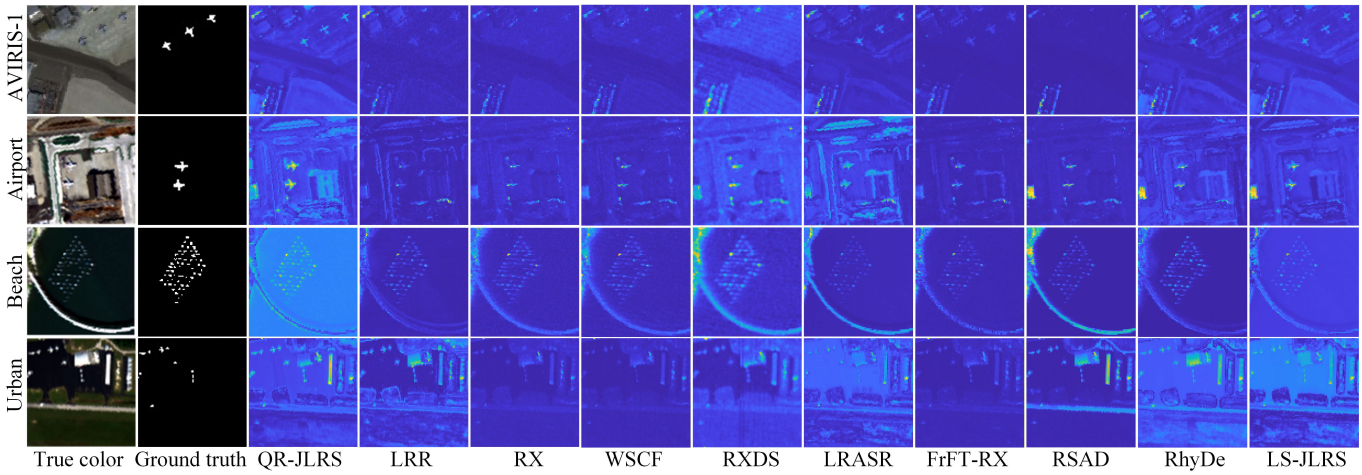


Fig. 6. Anomaly detection heat maps of different methods on the four *noise-free* test datasets.

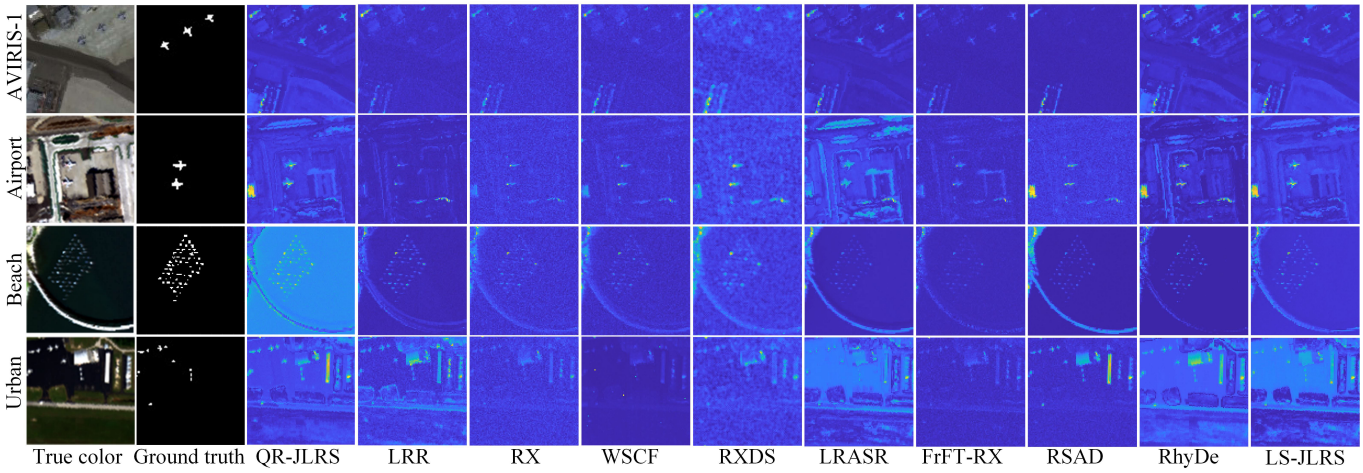


Fig. 7. Anomaly detection heat maps of different methods on four test datasets with *Gaussian noise*.

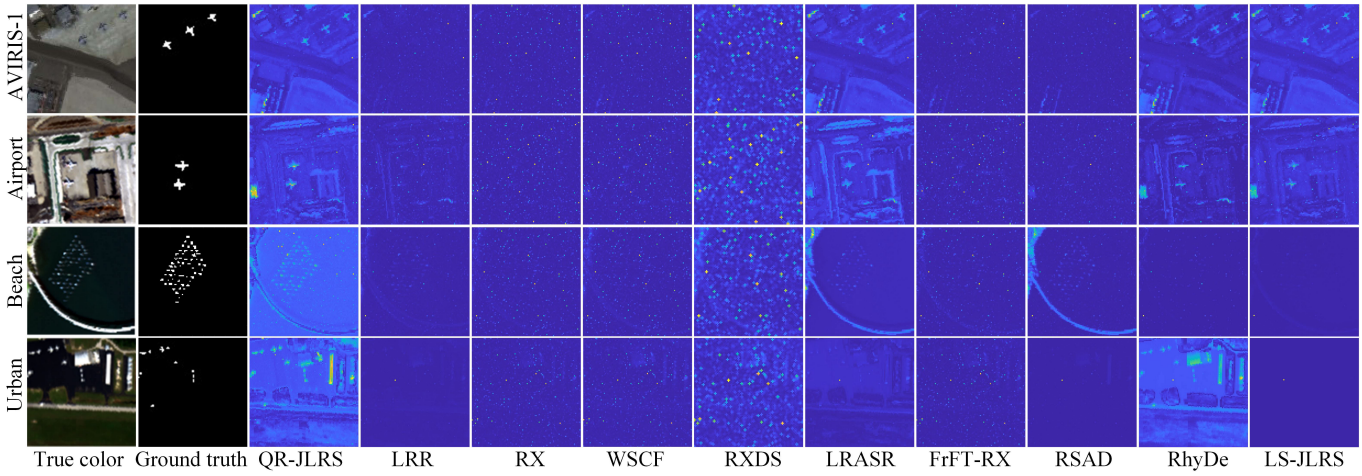


Fig. 8. Anomaly detection heat maps of different methods on four test datasets with *Student's t noise*.

F. Performance Evaluation via Separability Boxplots

To assess the separability between anomalies and the background across different noise conditions, we present the separability boxplots for all ten competing methods in Figs. 14–17. For each algorithm, a red box represents the distribution of detection scores for anomaly pixels, while a blue box represents the distribution of detection scores for background pixels. The central line in each box indicates the median score, the box spans from the 10th to the 90th

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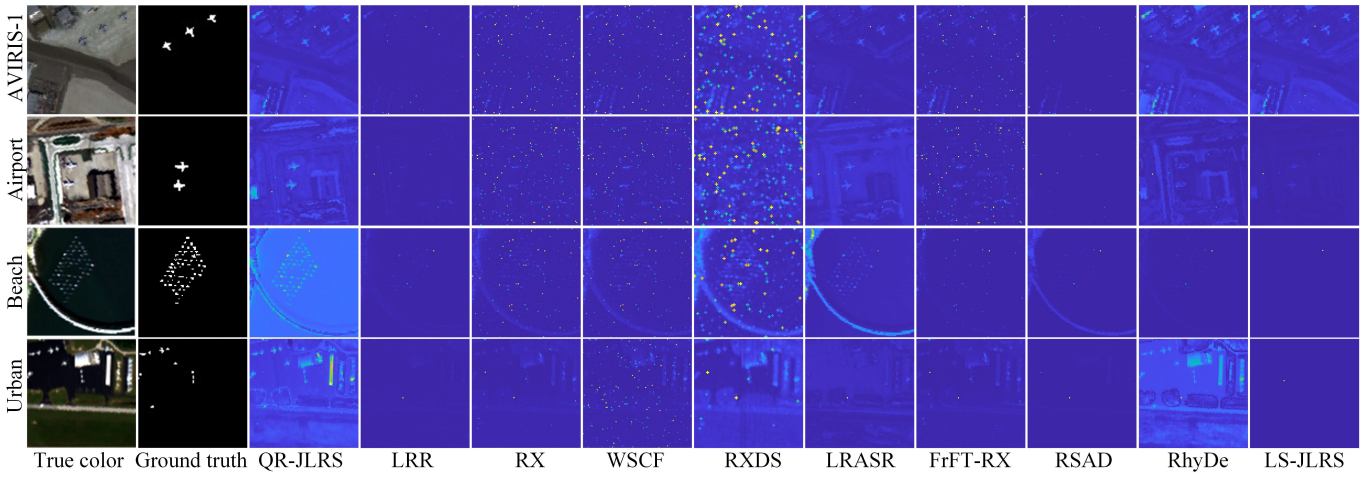


Fig. 9. Anomaly detection heat maps of different methods on four test datasets with *Cauchy noise*.

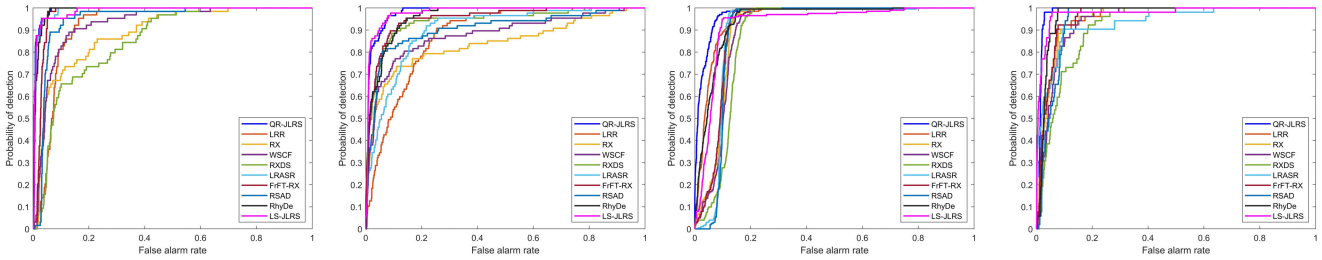


Fig. 10. ROC curves of the proposed QR-JLRS and the nine benchmark algorithms on four hyperspectral datasets under the *noise-free* condition. (a) AVIRIS-1. (b) Airport. (c) Beach. (d) Urban.

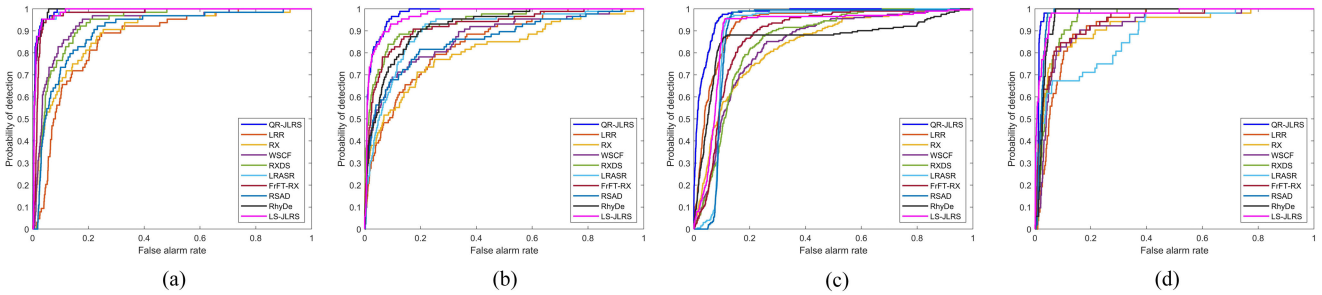


Fig. 11. ROC curves of the proposed QR-JLRS and the nine benchmark algorithms on four hyperspectral datasets contaminated with *Gaussian noise*. (a) AVIRIS-1. (b) Airport. (c) Beach. (d) Urban.

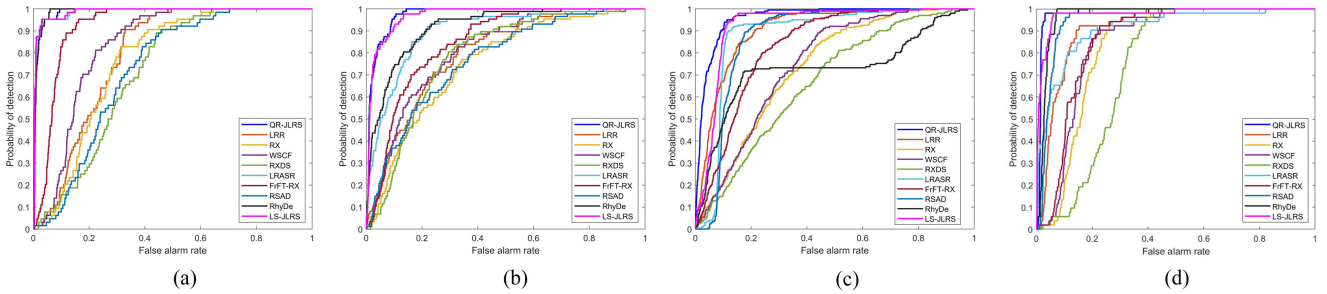
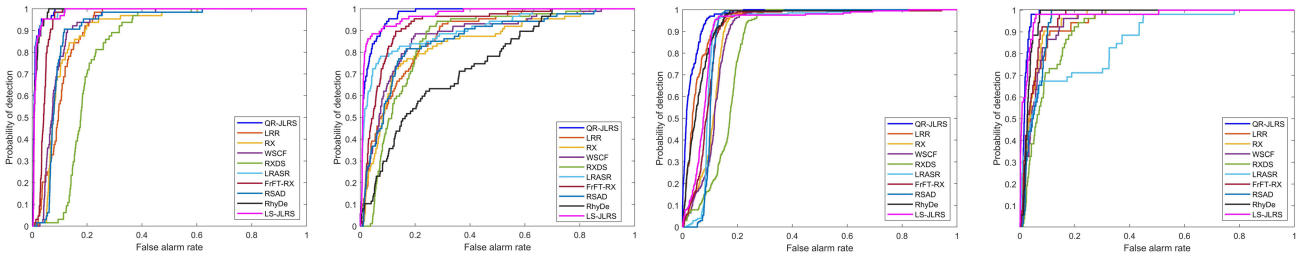


Fig. 12. ROC curves of the proposed QR-JLRS and the nine benchmark algorithms on four hyperspectral datasets under *Student's t noise*. (a) AVIRIS-1. (b) Airport. (c) Beach. (d) Urban.

percentile, and the whiskers extend to the full data range. A larger separation between the two boxes reflects a better discriminative ability of the detector.

In the noiseless case in Fig. 14, most methods exhibit a clear separation between anomaly and background boxes, with QR-JLRS showing one of the largest gaps across all

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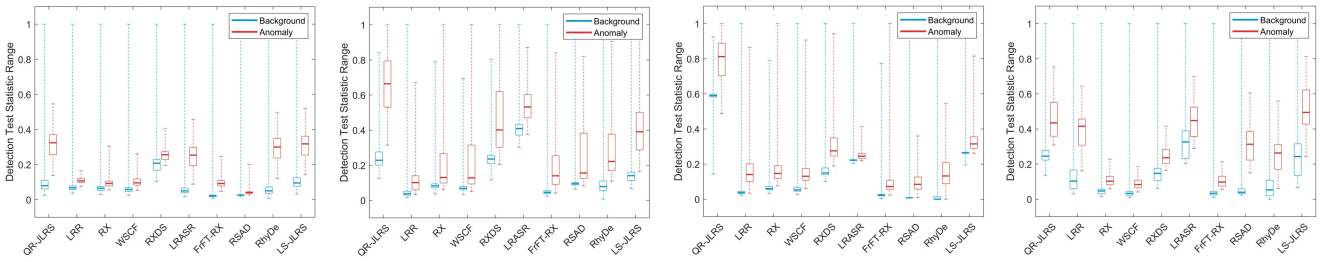
(a) (b) (c) (d)

Fig. 13. ROC curves of the proposed QR-JLRS and the nine benchmark algorithms on four hyperspectral datasets under *Cauchy noise*. (a) AVIRIS-1. (b) Airport. (c) Beach. (d) Urban.

TABLE I

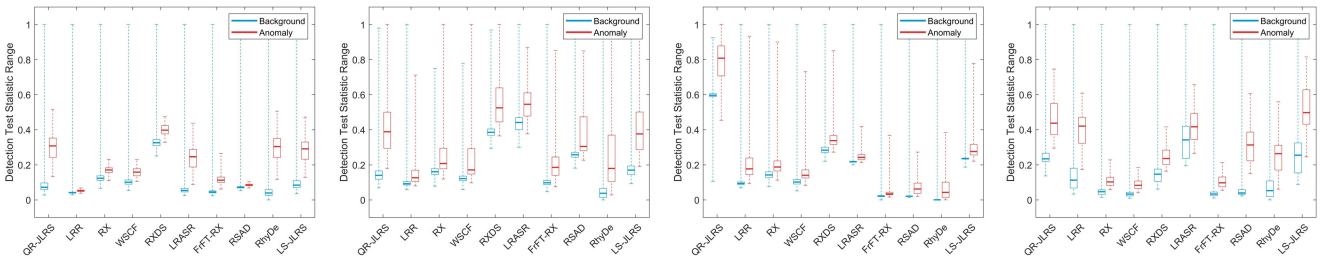
AUC SCORES ON THE FOUR DATASETS

Dataset	AVIRIS-1				Airport				Beach				Urban			
Noise Type	No noise	Gaussian	*t*	Cauchy	No noise	Gaussian	*t*	Cauchy	No noise	Gaussian	*t*	Cauchy	No noise	Gaussian	*t*	Cauchy
QR-JLRS	**0.9908**	**0.9892**	**0.9892**	**0.9882**	**0.9862**	**0.9777**	**0.9743**	**0.9735**	**0.9729**	**0.9719**	**0.9606**	**0.9719**	**0.9843**	**0.9838**	**0.9833**	**0.9805**
LRR	0.9230	0.8596	0.7923	0.8988	0.8818	0.8443	0.7948	0.8738	0.9498	0.9342	0.9057	0.9457	0.9536	0.9203	0.9150	0.9381
RX	0.8866	0.8867	0.7669	0.8922	0.8404	0.8139	0.7537	0.8406	0.9106	0.8363	0.7369	0.8960	0.9513	0.9186	0.8305	0.9506
WSCF	0.9254	0.9308	0.8347	0.9124	0.8803	0.8687	0.8093	0.8733	0.9014	0.8183	0.7510	0.8869	0.9413	0.9185	0.8511	0.9400
RXDS	0.8513	0.9267	0.7168	0.8047	0.9400	0.9387	0.7879	0.8501	0.8810	0.8540	0.6705	0.8419	0.9158	0.9492	0.7465	0.9123
LRASR	0.9870	0.9860	0.9848	0.9859	0.9081	0.9015	0.8878	0.8990	0.9035	0.9017	0.8978	0.8840	0.9352	0.8805	0.9132	0.8600
FrFT-RX	0.9711	0.9747	0.9242	0.9556	0.9533	0.9235	0.8456	0.9239	0.9147	0.8799	0.8328	0.9098	0.9584	0.9385	0.8655	0.9572
RSAD	0.9458	0.8974	0.7269	0.9036	0.9057	0.8603	0.7683	0.8568	0.8995	0.9011	0.8747	0.8979	0.9462	0.9766	0.9529	0.9458
RhyDe	0.9868	0.9881	0.9876	0.9863	0.9588	0.9195	0.9171	0.7451	0.9447	0.8673	0.7309	0.9415	0.9689	0.9696	0.9696	0.9694
LS-JLRS	0.9857	0.9867	0.9872	0.9845	0.9811	0.9720	0.9704	0.9705	0.9295	0.9074	0.9265	0.9158	0.9768	0.9753	0.9755	0.9742



(a) (b) (c) (d)

Fig. 14. Separability boxplots of the proposed QR-JLRS and the nine benchmark algorithms on four hyperspectral datasets under the *noise-free* condition. (a) AVIRIS-1. (b) Airport. (c) Beach. (d) Urban.



(a) (b) (c) (d)

Fig. 15. Separability boxplots of the proposed QR-JLRS and the nine benchmark algorithms on four hyperspectral datasets under *Gaussian noise*. (a) AVIRIS-1. (b) Airport. (c) Beach. (d) Urban.

four datasets, indicating its excellent separation ability. Under Gaussian noise in Fig. 15, the box overlap of some traditional methods (e.g., RX, LRR) increases to varying degrees, especially on the beach dataset, whereas QR-JLRS maintains a consistently wide gap, confirming its robustness to Gaussian perturbations.

More importantly, under heavy-tailed noise conditions in Figs. 16 and 17, the superiority of QR-JLRS becomes even

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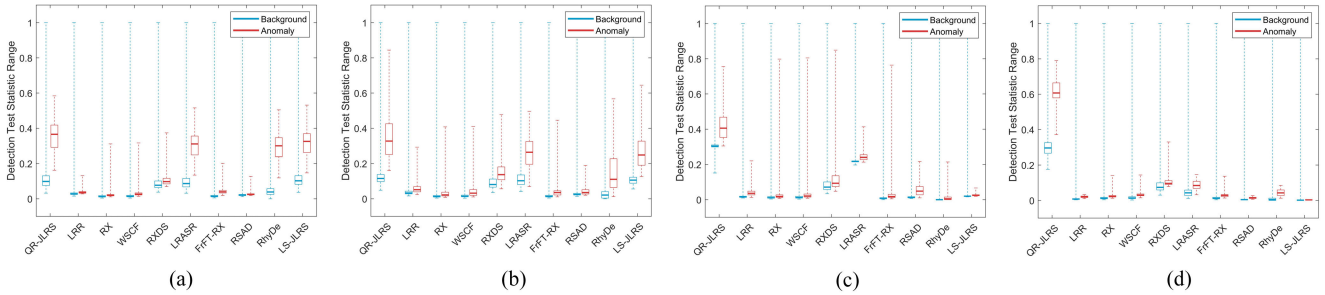


Fig. 16. Separability boxplots of the proposed QR-JLRS and the nine benchmark algorithms on four hyperspectral datasets under *Student's t noise*. (a) AVIRIS-1. (b) Airport. (c) Beach. (d) Urban.

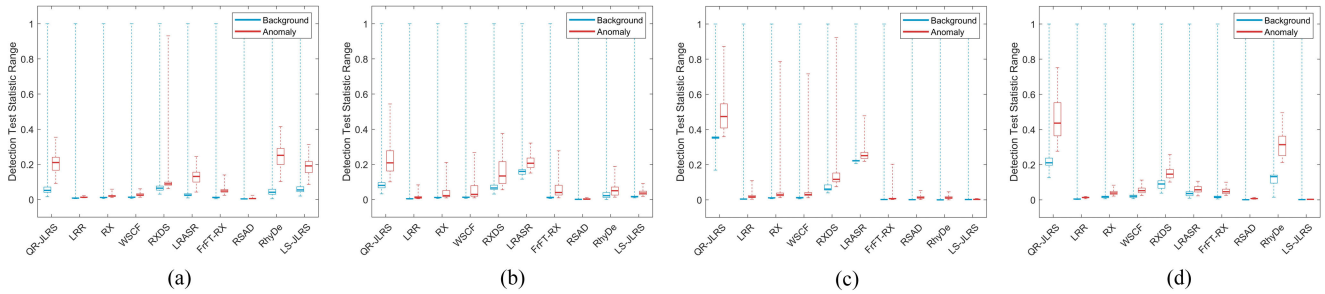


Fig. 17. Separability boxplots of the proposed QR-JLRS and the nine benchmark algorithms on four hyperspectral datasets under *Cauchy noise*. (a) AVIRIS-1. (b) Airport. (c) Beach. (d) Urban.

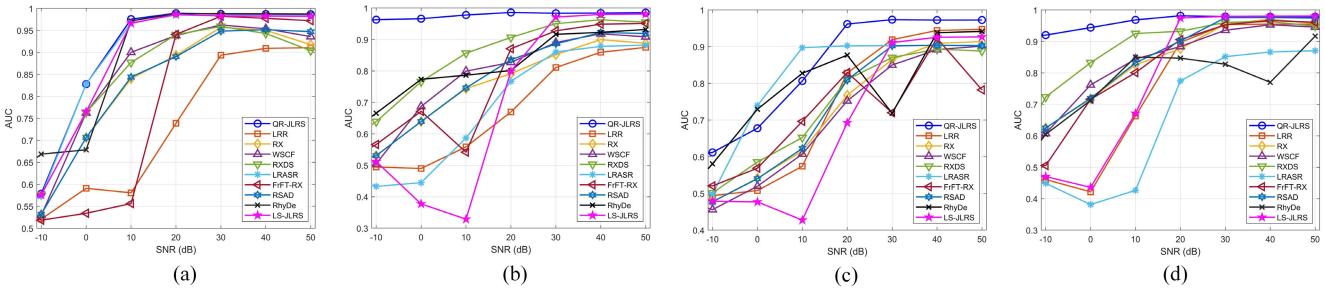


Fig. 18. AUC performance comparison of the proposed QR-JLRS and the nine benchmark algorithms under various SNR levels on four hyperspectral datasets contaminated by *Gaussian noise*. (a) AVIRIS-1. (b) Airport. (c) Beach. (d) Urban.

more evident. For instance, on the airport and beach datasets under t-distribution noise and Cauchy noise, the anomaly and background boxes for methods like LRR and RhyDe show substantial overlap, indicating poor separation. Furthermore, under these heavy-tailed noise conditions, the LS-JLRS method exhibits almost complete overlap between background and anomaly boxes on the beach and urban datasets, indicating a severe loss of discriminative capability. In contrast, QR-JLRS maintains a distinct separation with the anomaly box positioned at significantly higher score values. This pattern is consistently observed across all datasets and both heavy-tailed noise types, demonstrating that the QR formulation in QR-JLRS effectively preserves discriminative power even when the data contains outliers.

The separability boxplots analysis quantitatively supports that QR-JLRS provides not only higher detection scores for anomalies but also better suppression of background responses, leading to superior separability that is maintained across a variety of noise conditions, particularly in challenging heavy-tailed noise environments.

G. AUC Performance Versus SNR

To more intuitively evaluate the robustness of different detection algorithms under varying noise levels, we plot the AUC values as a function of the SNR for Gaussian, Student's t, and Cauchy noise in Figs. 18–20, respectively. As expected, for all three noise types, the detection performance of all methods generally improves as the SNR increases. Under Gaussian noise in Fig. 18, QR-JLRS maintains a clear performance advantage over most competing methods across a broad SNR range.

Under Student's t noise in Fig. 19 and Cauchy noise in Fig. 20, the performance gap becomes more pronounced. In the SNR range of 10–40 dB, which is typical for real HSI scenarios, QR-JLRS consistently achieves the highest

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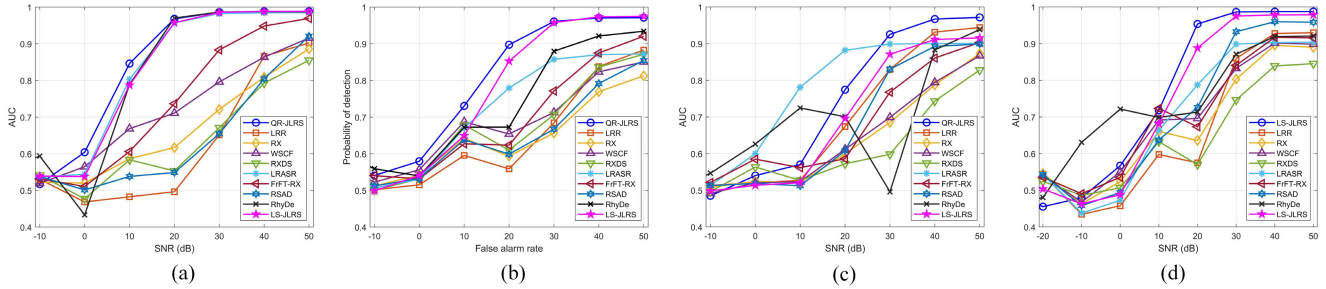


Fig. 19. AUC performance comparison of the proposed QR-JLRS and the nine benchmark algorithms under various SNR levels on four hyperspectral datasets contaminated by *Student's t noise*. (a) AVIRIS-1. (b) Airport. (c) Beach. (d) Urban.

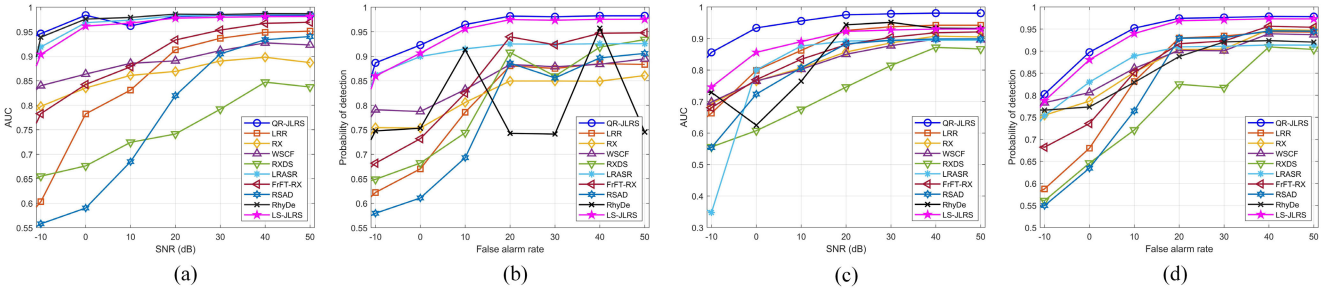


Fig. 20. AUC performance comparison of the proposed QR-JLRS and the nine benchmark algorithms under various SNR levels on four hyperspectral datasets contaminated by *Cauchy noise*. (a) AVIRIS-1. (b) Airport. (c) Beach. (d) Urban.

TABLE II
RUNNING TIME (S) OF DIFFERENT ALGORITHMS ON FOUR DATASETS

Algorithms	QR-JLRS	LRR	RX	WSCF	RXDS	LRASR	FrDT-RX	RSAD	RhyDe	LS-JLRS
AVIRIS-1	9.149	38.179	0.035	0.036	0.390	17.952	9.519	15.351	0.093	9.405
Airport	11.368	48.668	0.049	0.052	0.610	26.659	13.850	15.593	0.111	10.789
Beach	9.583	42.135	0.041	0.039	0.427	24.084	11.178	38.562	0.098	9.458
Urban	11.508	47.451	0.043	0.054	0.458	26.282	11.193	33.277	0.106	10.419

or near-highest AUC values across all tested datasets. This behavior underscores the effectiveness of the QR formulation in handling heavy-tailed noise, which is often encountered in practical remote sensing environments.

Overall, the SNR-based analysis confirms that QR-JLRS is not only effective under fixed noise conditions but also maintains superior robustness across a wide range of noise intensities, making it a reliable choice for real-world HAD applications.

H. Computational Complexity and Running Time Analysis

In this section, we analyze the computational complexity of the proposed preconditioned QR-JLRS algorithm and further validate its practical efficiency using running time comparisons. The per-iteration computational complexity is analyzed based on the primal-dual updates in (41)–(43). The dominant computational cost in the primal update of $\boldsymbol{S}$, as shown in (41a), arises from the matrix multiplication $\boldsymbol{D}^\top \boldsymbol{Y}^{(\ell)}$, which requires $\mathcal{O}(LMN)$ operations. Similarly, the updates of $\boldsymbol{E}$ and $\boldsymbol{Q}_\tau^R$ in (41b) and (41c) mainly consist of matrix additions and element-wise operations, resulting in a computational complexity of $\mathcal{O}(LN)$.

The main computational burden arises from the proximal projection associated with the spectral norm constraint, which requires SVD. Specifically, for $\boldsymbol{Z}_1 \in \mathbb{R}^{M\times N}$, the complexity of SVD $\boldsymbol{Z}_1 = \boldsymbol{U}\text{diag}(\boldsymbol{\sigma})\boldsymbol{V}^\top$ is $\mathcal{O}(\min(MN^2, M^2N))$. In addition, the proximal operators for the ℓ_∞ and $\ell_{2,1}$ constraints in (28) and (29) are closed-form projections, and therefore introduce relatively minor computational cost. Therefore, the overall per-iteration computational complexity of the proposed preconditioned QR-JLRS can be summarized as: $\mathcal{O}(LMN) + \mathcal{O}(\min(MN^2, M^2N))$.

To further evaluate the practical computational efficiency, Table II reports the running time of different methods on four hyperspectral datasets under the noise-free setting. It can be observed that QR-JLRS is slower than classical detectors such as RX, WSCF, and RXDS, due to the inclusion of low-rank modeling and SVD operations. However, compared with representation-based methods such as LRR and LRASR, QR-JLRS generally achieves lower running time. This is mainly attributed to the faster convergence of the proposed preconditioned optimization scheme, which reduces the number of required iterations.

In summary, although the proposed algorithm achieves improved convergence performance via the preconditioned

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primal-dual framework, the involvement of SVD operations still introduces non-negligible computational overhead, especially when dealing with large-scale hyperspectral data. To address this issue, we will further improve the computational efficiency in future work to enhance the scalability of the algorithm for practical applications.

V. Conclusion

In this article, we propose a robust hyperspectral anomaly detector termed QR-JLRS, which integrates QR with LRASR to enhance resilience against heavy-tailed noise and outliers. To efficiently solve the resulting nonsmooth optimization problem, we design a PDHG algorithm and further improve its convergence stability and speed by incorporating a diagonal preconditioning strategy. Experimental results on multiple datasets under various noise conditions, including Gaussian and heavy-tailed noise, demonstrate that QR-JLRS maintains competitive detection accuracy while exhibiting excellent robustness, particularly in complex environments contaminated by heavy-tailed noise. An important direction for future work is to further improve the computational efficiency of the proposed method, especially for large-scale hyperspectral data.

Appendix
Derivation of the Proximal Operator in (27)

This appendix provides the detailed derivation of the proximal operator associated with the spectral norm constraint in (27).

The indicator function of the spectral norm ball $\mathbb{Z}_1$ is defined as

$$\mathcal{I}_{\mathbb{Z}_1}(\boldsymbol{Z}) = \begin{cases} 0, & \text{if } \|\boldsymbol{Z}\|_\sigma \le 1 \\ +\infty, & \text{otherwise.} \end{cases} \tag{45}$$

By definition, the proximal operator of $\mathcal{I}_{\mathbb{Z}_1}$ at $\boldsymbol{X}$ is given by

$$\text{prox}_{\mathcal{I}_{\mathbb{Z}_1}}(\boldsymbol{X}) = \arg\min_{\boldsymbol{Z}} \left\{ \mathcal{I}_{\mathbb{Z}_1}(\boldsymbol{Z}) + \frac{1}{2}\|\boldsymbol{Z}-\boldsymbol{X}\|_F^2 \right\}. \tag{46}$$

Note that $\eta > 0$ does not affect the proximal operator of an indicator function. Therefore, the parameter η appearing in (27) is omitted here for notational simplicity. The above problem is equivalent to the Euclidean projection of $\boldsymbol{X}$ onto the spectral norm ball, namely,

$$\text{prox}_{\mathcal{I}_{\mathbb{Z}_1}}(\boldsymbol{X}) = \arg\min_{\|\boldsymbol{Z}\|_\sigma \le 1} \frac{1}{2}\|\boldsymbol{Z}-\boldsymbol{X}\|_F^2. \tag{47}$$

Let the SVD of $\boldsymbol{X}$ be $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^\top$, where $\boldsymbol{\Sigma} = \text{diag}(\sigma_1, \sigma_2, \ldots, \sigma_k)$ with $\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_k \ge 0$. Due to the unitary invariance of the Frobenius norm, the optimal solution shares the same left and right singular vectors as $\boldsymbol{X}$. Accordingly, the problem reduces to the following optimization with respect to the singular values: $\min_{\hat{\sigma}_i \le 1} 1/2 \sum_{i=1}^{k} (\hat{\sigma}_i - \sigma_i)^2$. This problem is separable, and its optimal solution is given by $\hat{\sigma}_i = \min(\sigma_i, 1)$ for $i = 1, \ldots, k$. Substituting the optimal singular values back into the SVD reconstruction, we obtain directly the proximal operator for the spectral norm constraint reported in (27).

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